Light escape cones in the locally nonrotating reference frames of the Kerr–de Sitter superspinars and related superspinar shadows

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The construction of local escape cones and their complementary cones, related to the locally nonrotating frames (LNRFs) orbiting superspinars with the external field described by the Kerr–de Sitter (KdS) naked singularity (NS) spacetimes, is described for all possible classes of the KdS NS spacetimes and all possible positions of the photon emitter. The notion of the local escape cones is then applied to construct the shadow of the KdS superspinars related to the distant static observers represented by the LNRFs located near the so-called static radius where the spacetime is close to the asymptotically flat region of the Kerr spacetimes. The shadow construction is focused on the KdS superspinars limited by the cosmological and astrophysical restrictions on the spacetime parameters, namely, the superspinar dimensionless spin and the dimensionless cosmological parameter relating the cosmological constant and the superspinar's mass.

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I. INTRODUCTION

Event Horizon Telescope observations of the central regions of the galaxy M87 [1–9] and our Galaxy [source Sagittarius (Sgr) A*] [10–16] have opened an intensive study of the black hole (BH) shadow and the physics of processes close to the BH horizon. The observations are consistent with the expectation that the central object in both observed sources is a supermassive Kerr black hole. However, the Kerr geometry can also describe Kerr naked singularities (NSs), which contain a region of causality violation.

The Penrose cosmic censorship hypothesis forbids the existence of NSs, however, the hypothesis was not proved, and the validity of the weak cosmic censorship conjecture, which forbids the appearance of NSs clothing each one in an event horizon [17], is still an open question [18,19]. A variety of realistic NS spacetimes have been found—it has been shown that gravitational collapse can lead to NSs [20–24].[1]

Therefore, the properties of Kerr NS spacetimes have been studied intensively in [25–27], while the optical phenomena related to the Kerr NS spacetimes were studied in [28–30] or for their generalizations in [31].

The optical phenomena around Kerr-Newman BHs and NSs, along with the braneworld Kerr BHs and NSs, have been studied in [32–41], and for the regular BHs [42–45] constructed in the framework of general relativity in combination with various variants of nonlinear electrodynamics, these phenomena have been treated, e.g., in [46–48]. Gravitational lensing by spherically symmetric static naked singularities modeled in the framework of the Einstein massless scalar field theory has been studied in [49–52]. It has been shown that naked singularities can be observationally differentiated from black holes.

A modification of the Kerr NS spacetimes that does not exhibit the region of causality violations (time machine, see [53]), located in the vicinity of the ring singularity, has been proposed, being based on the stringy solution corresponding to rapidly rotating compact objects, so-called Kerr superspinars, where the interior is assumed to be governed by the string theory, and the exterior is described by the standard Kerr NS geometry [54]. The pathological time machine region is then covered by correctly behaving "stringy region," which is expected to have a boundary at radius $\mathcal{R} > 0$. The Kerr superspinars can be considered as primordial, being remnants of the initial stringy phase of the evolution of the Universe [54,55]. Their properties have been extensively studied, yielding some extraordinary results [56–59].

The optical phenomena associated with the Kerr superspinars have been studied, e.g., in [30,60]—it was shown that the superspinar shadow depends on the edge at $r = \mathcal{R}$, where the external Kerr spacetime and the internal spacetime join each other and is significantly different from the BH shadow.

It is worth studying in detail the construction of the shadow of both the complete Kerr NS spacetimes and the Kerr superspinars external geometry, and in their extensions, to test their observational relevance, since we can expect that in the near future, due to the rapid development of the observational and data analysis techniques, we will be able to distinguish very subtle details of the compact

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[1]The big bang itself represents a NS.

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objects, thus obtaining important information on various physical phenomena.

Because of the cosmological observations related to both the cosmic microwave background measurements and supernova tests that indicate the existence of dark energy with properties very close to the vacuum energy represented by the cosmological constant [61,62], it is important to clarify the role of the cosmological constant Λ in astrophysical phenomena related to Kerr superspinars. The role of Λ in accretion processes in active galactic nuclei has been discussed in [63–66] and the motion of the gravitationally bound galaxies in [67]. The optical phenomena and the BH shadow have been studied for Kerr–de Sitter (KdS) BHs in [68], namely, for locally nonrotating reference frames (LNRFs), frames related to circular geodesic orbits, and for radially freely falling/escaping (geodesic) frames. For an alternative KdS BH solution and its shadow see [69,70]. Furthermore, the influence of the observer's state of motion on the BH shadow, related to that of the Sgr A* BH, is described in [71]. The shadows of BHs embedded in the expanding de Sitter universe have been studied in [72–74], and for charged and rotating BHs in [75], while the connection with the Friedman-Lemaître-Robertson-Walker cosmological metric has been considered in [76]. For Kerr–de Sitter NSs the shadow was treated, with inclusion of the region of negative radii, in [77], considering LNRFs, and radially freely falling/escaping (geodesic) frames.

In the present paper we focus attention on the shadow of the KdS superspinars, whose surface at $r = \text{const} = \mathcal{R} > 0$ functions similar to the BH horizon, thus excluding the possibility of reaching the causality violation region. We discuss in detail the construction of the light escape cones (LECs) related to the LNRFs, which are important from an astrophysical point of view [66]. The LECs are then applied in the construction of the KdS superspinar's shadow.

Due to the variability and complexity of the KdS NS spacetimes, which represent the exterior of the KdS superspinars, we restrict our study of the shadows to the class of KdS spacetimes whose parameters are allowed by the observational constraints. We use the relict cosmological constant $\Lambda \sim 10^{-52} m^{-2}$ implied by cosmic observations [78,79] and the constraints on the mass and spin of the compact objects imposed by observations of active galactic nuclei in large galaxies or even in the galactic clusters [65,80]. This class of the KdS NS spacetimes allows the existence of static radii [77,81,82]—the spacetime at (and around) this radius represents in the KdS spacetimes the best local approximation to the asymptotically flat region of the Kerr spacetimes [82,83]. Furthermore, the static radius also represents an upper limit on the extension of gravitationally bound systems (e.g., the dark matter halos) in spacetimes with the relict cosmological constant [80,84]. At the static radius, the LNRF observers approximate the geodetic stationary observers at an unstable equilibrium point where the gravitational attraction of the BH or superspinar is just balanced by the cosmic repulsion.

The concept of LECs [and complementary light capture cones (LCCs)] enables construction of the KdS superspinar shadow (silhouette) in the same way as in the case of the BH or NS shadows [30,39,68,77], with respect to any observer, including those orbiting very close to the central object. Here, we restrict our attention to the standard case of a shadow (silhouette) cast by a KdS superspinar located between a very distant observer and an illuminating source much larger than the compact object [85]. To construct the LEC and the shadow, we assume the light source at the position of the observer and determine the motion of the photons emitted in different directions. The results obtained for the construction of the LEC are then inverted to obtain the shadow (for details see [77]). A special class of null geodesics is considered in the analysis, namely, null geodesics corresponding to spherical photon orbits, which are now considered only at the radii corresponding to the external spacetime of the superspinars. The construction of the LEC and the complementary LCC for the superspinars is compared with the construction in the NS case.

II. KERR–DE SITTER SUPERSPINAR (NS) GEOMETRY

We assume that, similar to the pure Kerr case, the exterior of the Kerr–de Sitter superspinar spacetimes are determined by the standard Kerr–de Sitter naked singularity parameters, namely, the superspinar mass M, its spin J, which is replaced in a standard way by the dimensionless rotational parameter $a = J/M^2 > 1$, and the cosmological constant Λ, which is replaced by the dimensionless cosmological parameter $y = \frac{1}{3}\Lambda M^2$. Another parameter $\mathcal{R}$ is important for the description of the Kerr–de Sitter superspinar, as it determines its boundary, separating its interior, which is assumed to be described by the string theory, from the exterior, where the stringy effects are assumed to be irrelevant and the spacetime is described by the standard Kerr–de Sitter geometry. The value of the boundary parameter $\mathcal{R}$ can be set arbitrarily, at $\mathcal{R} = r = \text{const} > 0$; usually it is assumed at $\mathcal{R} = 0$ or slightly above.

As usual, we use the Boyer-Lindquist spheroidal coordinates t, r, θ, ϕ with geometric system of units ($c = G = 1$). We also introduce the dimensionless line element and the coordinates $s/M \to s$, $t/M \to t$, $r/M \to r$. This is equivalent to putting $M = 1$. The line element then reads

$$\begin{aligned} \mathrm{d}s^2 = {} & \frac{a^2 \sin^2\theta \Delta_\theta - \Delta_r}{I^2 \rho^2} \mathrm{d}t^2 + \frac{\rho^2}{\Delta_r} \mathrm{d}r^2 + \frac{\rho^2}{\Delta_\theta} \mathrm{d}\theta^2 \\ & + \frac{\sin^2\theta}{I^2 \rho^2} A \mathrm{d}\phi^2 + \frac{2a \sin^2\theta}{I^2 \rho^2} [\Delta_r - \Delta_\theta (r^2 + a^2)] \mathrm{d}t \mathrm{d}\phi, \end{aligned} \tag{1}$$

where

$$\Delta_r = (1 - yr^2)(r^2 + a^2) - 2r, \tag{2}$$

$$\Delta_\theta = 1 + a^2 y \cos^2\theta, \tag{3}$$

$$I = 1 + a^2 y, \tag{4}$$

$$\rho^2 = r^2 + a^2 \cos^2\theta, \tag{5}$$

and

$$A = (r^2 + a^2)^2 \Delta_\theta - a^2 \Delta_r \sin^2\theta. \tag{6}$$

In some special cases, such as the situation in the vicinity of the spacetime ring singularity, it is convenient to construct the figures in the so-called Kerr-Schild coordinates, which are related to the Boyer-Lindquist coordinates by the relations

$$x^2 + y^2 = (r^2 + a^2)\sin^2\theta,$$
$$z^2 = r^2 \cos^2\theta. \tag{7}$$

Note that in the relations (7) y denotes a Cartesian coordinate, not a cosmological parameter.

A. Event horizons

The roots of the quartic equation

$$\Delta_r = 0, \tag{8}$$

which can be rewritten in the form

$$y = y_h(r; a^2) \equiv \frac{r^2 - 2r + a^2}{r^2(r^2 + a^2)}, \tag{9}$$

determine the coordinate pseudosingularities of the KdS geometry. Following the notation used in our previous paper [77], we denote them as r_c^-, r_i, r_o, r_c, where $r_c^- < 0 < r_i < r_o < r_c$, corresponding to the cosmological horizon in the negative radius region, the inner BH (Cauchy) horizon, the outer BH horizon, and the cosmological horizon. The number and position of the horizons for a given parameter y_0 can be illustrated by the intersection of the horizontal y_0 line with the graph of the function $y_h(r; a^2)$, whose behavior is shown in Fig. 1(a) for different values of spin parameter a^2. It can be observed that the cosmological horizon r_c^- in region of negative radii exists for any positive values of $y > 0$ and $a^2 > 0$. However, we have to stress that in our study in this paper we consider only the KdS NS spacetimes excluding the region of $r < 0$, since it is assumed to be irrelevant because of the assumption $\mathcal{R} \geq 0$. Thus, it is clear that only the cosmological horizon in the region of $r > 0$ is relevant in our study. The number and nature of horizons in the $r > 0$ region, which we discuss separately below, depends on the value of y_0 compared to the extremes $y_{\min(h)}(a_0^2)$, $y_{\max(h)}(a_0^2)$, for a given spin parameter $a_0^2 > 0$. Here and in the following enumeration the functions $y_{\min/\max(h)}(a^2)$ describe the minima/maxima of the function $y_h(r; a^2)$.

(1) Case $0 \leq a_0^2 \leq 1$.

The function $y_h(r; a_0^2)$ has a positive maximum $y_{\max(h)}(a_0^2) > 0$ and nonpositive minimum $y_{\min(h)}(a_0^2) \leq 0$ [see the gray, red, and blue curves in Fig. 1(a)]; for $a_0^2 = 0$ the function $y_h(r; a_0^2)$ has no minimum, since $y_h(r; 0) \to -\infty$ as $r \to 0^+$, while $y_{\max(h)}(a_0^2) = y_{crit(SdS)} \equiv 1/27 \doteq 0.03704$ [see the gray curve in Fig. 1(a)]; $y_{crit(SdS)}$ denotes the critical value of the cosmological parameter that allows the Schwarzschild–de Sitter BH spacetimes to exist. For $a_0^2 = 1$, $y_{\min(h)}(a_0^2) = 0$ [see the blue curve in Fig. 1(a)].

(i) $0 < y_0 < y_{\max(h)}(a_0^2)$.
The geometry describes a BH spacetime; the radii $0 < r_i < r_o < r_c$ correspond, respectively, to the inner BH horizon, the outer BH horizon, and the cosmological horizon; the spacetime is dynamic in regions $r_i < r < r_o$.

(ii) $y_0 = y_{\max(h)}(a_0^2)$.
Marginal NS spacetimes, where the outer BH and the cosmological horizons merge at $r_o = r_c$.

(iii) $y_{\max(h)}(a_0^2) < y_0$.
The geometry describes the NS spacetimes with one cosmological horizon $0 < r_c$ in the region of positive radii.

(2) Case $1 < a_0^2 \leq a_{\text{crit}}^2 \equiv 3(3 + 2\sqrt{3})/16 \doteq 1.21202$.

Both local extrema $y_{\min(h)}(a_0^2)$ and $y_{\max(h)}(a_0^2)$ of the function $y_h(r; a_0^2)$ are positive, $0 < y_{\min(h)}(a_0^2) < y_{\max(h)}(a_0^2) \leq y_{crit(KdS)} \equiv 16(26\sqrt{3} - 45)/9 \doteq 0.05924$ [see the green curve in Fig. 1(a)]—these extrema merge for $a_0^2 = a_{\text{crit}}^2$ at the inflection point in $y = y_{crit(KdS)}$ [see the orange curve in Fig. 1(a)]; $y_{crit(KdS)}$ and a_{crit}^2 have the meaning of the critical values of the cosmological and spin parameters, respectively, that still allow the existence of Kerr–de Sitter BH spacetimes.

(i) $y_0 < y_{\min(h)}(a_0^2)$ or $y_{\max(h)}(a_0^2) < y_0$.
The geometry describes a NS spacetime with properties analogous to the previous NS case.

(ii) $y_{\min(h)}(a_0^2) < y_0 < y_{\max(h)}(a_0^2)$.
Geometry describes the BH spacetimes with properties analogous to the previous BH case.

(iii) $y_0 = y_{\min(h)}(a_0^2)$.
Limit case of the extreme BH spacetimes with a merging inner and outer BH horizon.

(iv) $y_0 = y_{\max(h)}(a_0^2)$.
Limit case of the marginal NS spacetimes with properties analogous to the previous marginal NS case.

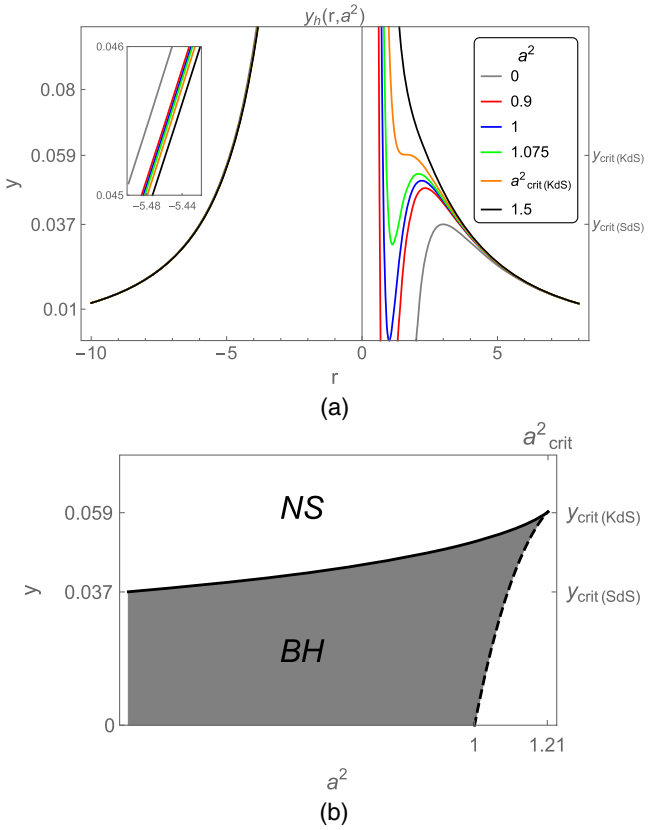


FIG. 1. (a) Behavior of the function $y_h(r;a^2)$ for typical values of the spin parameter a^2. For a given cosmological parameter y_0, the intersections of the horizontal line $y=y_0$ with the curves $y=y_h(r;a^2)$ give the number and locations of the event horizons of the corresponding spacetime. (b) Behavior of the functions $y_{\max(h)}(a^2)$ (full black curve) and $y_{\min(h)}(a^2)$ (dashed black curve) and the induced partitioning of the parameter plane (a^2-y) into regions corresponding to the BH spacetimes (gray region) and NS spacetimes (white region).

(v) $a_0^2=a_{\rm crit}^2$, $y_0=y_{crit(KdS)}$.
Extreme case of geometry that describes the ultraextreme NS spacetime with $r_i=r_0=r_c$.

(3) Case $a_0^2>a_{crit(KdS)}^2$.
There are no local extrema of the function $y_h(r;a_0^2)$ [see the black curve in Fig. 1(a)], and for any value of y_0 the geometry corresponds to the NS spacetimes with properties analogous to the previous NS cases.

The behavior of the functions $y_{\min/\max(h)}(a^2)$ is shown in Fig. 1(b). They can be expressed as follows:

$$y_{\min(h)}(a^2)\equiv\frac{1}{12}\left[X-\frac{\sqrt{6}}{a^6}\sqrt{Y+Z}+\sqrt{3}z\right], \quad (10)$$

and

$$y_{\max(h)}(a^2)\equiv\begin{cases} y^{-}_{\max(h)}(a^2), & \text{for } 0\le a^2<a^2_{\mp(h)};\\ y^{+}_{\max(h)}(a^2), & \text{for } a^2_{\mp(h)}\le a^2<a^2_{\rm crit};\end{cases} \quad (11)$$

where $a^2_{\mp(h)}\equiv\frac{1}{48}(3+2\sqrt{3})=0.135$, $a^2_{\rm crit}\equiv\frac{3}{16}(3+2\sqrt{3})=1.212$ is the maximum spin parameter allowing the existence of the BH spacetimes,

$$y^{\pm}_{\max(h)}(a^2)\equiv\frac{1}{12}\left[X+\frac{\sqrt{6}}{a^6}\sqrt{Y\pm Z}\pm\sqrt{3}z\right], \quad (12)$$

$$X=-\frac{3(1+4a^2)}{a^4},$$

$$Y=3a^4(1-80a^2)-2\times3^{1/3}u^{1/3}a^2-\frac{2\times3^{2/3}p_+a^8}{u^{1/3}},$$

$$Z=\frac{3\sqrt{3}(768a^4-96a^2-1)}{z}, \quad (13)$$

$$z=\sqrt{\frac{u^{1/3}(4\times3^{1/3}u^{1/3}+3a^2-240a^4)+4\times3^{2/3}p_+a^6}{u^{1/3}a^{10}}}, \quad (14)$$

$$u=1944a^{10}+18432a^{14}+\sqrt{3}\sqrt{-p_-^3a^{18}}, \quad (15)$$

and

$$p_\pm=256a^4\pm288a^2-27. \quad (16)$$

The exterior of KdS superspinar spacetimes is thus described by the KdS NS geometry. In the (y-a^2) plane, the parameters y, a^2 of such spacetimes correspond to the region shown in white in Fig. 1(b).

Since our intention here is to deal with astrophysically relevant objects, we admit that the mass parameter M of the central object can be as large as $M\sim10^{11}M_\odot$, as suggested by the case of the most massive BH observed so far, with mass of $M=7.10^{10}M_\odot$ in the source Tonantzintla 618. Here $M_\odot=1.5\times10^3m$ is the mass of the Sun in geometric units. Considering the current cosmological observations, which imply a very small value for the cosmological constant, $\Lambda\sim1.1\times10^{-52}m^{-2}$, we find that the realistic estimates of the cosmological parameter parameter y are at most $y\sim10^{-34}$. However, for the sake of illustration, we consider here values of the cosmological parameter y, which are many orders of magnitude higher than the realistic ones, but which still correspond to class IV KdS spacetimes, according to the classification introduced in [28], which are the main object of our investigation and which give results qualitatively corresponding to the realistic values of y.

The parameters of the KdS spacetimes of class IV correspond to the case $y<y_{\min(h)}(a^2)$ and we select

parameters giving clear demonstration of the relevant optical phenomena.

B. Superspinar boundary surface

We assume, in accordance with other related papers [30,57,58,60], that the boundary surface is formed by a sphere of constant radius $r(\theta) = \mathcal{R}$ with properties similar to the BH event horizon. Although considerations that do not rule out the possibility of nonzero emissivity or reflectivity can be found in the literature [86], in the present paper we regard the superspinar surface acting as a one-way membrane, i.e., it absorbs all incoming radiation and matter and does not radiate itself. Therefore, we demand that photons with radial turning point at $r = r_t$ can be detected by a distant observer if $r_t > \mathcal{R}$. We assume that the stringy effects are appreciable in the region where the curvature of the spacetime is comparable to the Planck scale, hence we fix the superspinar surface $\mathcal{R} \sim 0$ for our study.

It is well known [87] that the complete Kerr NS spacetimes contain the causality violation region, where the axial Killing vector $\xi_{(\phi)} = \partial/\partial\phi$ becomes timelike and hence in principle allows the existence of closed timelike worldlines. As proved in [88], such pathological behavior persists in the presence of the cosmological constant in one of the special parts of the Kerr–de Sitter spacetimes, referred to as Carter's blocks [89], which contain the ring singularity. It is shown that in this block there exists a nonempty region determined by

$$\xi^{\mu}_{(\phi)}\xi_{(\phi)\mu} = g_{\phi\phi} = \frac{\sin^2\theta}{I^2\rho^2}A < 0, \tag{17}$$

which acts as a so-called Carter's time machine, that generates closed timelike curves that can be extended through any point of the appropriate Carter's block. The analysis of the function A defined in Eq. (6) shows that the condition (17) can be met at small negative radii $r < 0$ in the vicinity of the ring singularity.

C. Ergosphere

The ergosphere region is given by the inequality

$$g_{tt} \equiv \frac{a^2\sin^2\theta\Delta_\theta - \Delta_r}{I^2\rho^2} \geq 0, \tag{18}$$

where equality corresponds to the ergosphere boundary, i.e., the ergosurface. In the equatorial plane the radii of the ergosurface are given by the condition

$$a^2 - \Delta_r = 0, \tag{19}$$

which is a quartic equation that always has a solution at $r = 0$, and in the case of the KdS spacetimes with $a^2y < 1$ that we are dealing with in this paper, it has three other real roots $r^{(-)}_{\rm erg}$, $r^{\pm}_{\rm erg}$, where $r^{(-)}_{\rm erg} < 0 < r^{+}_{\rm erg} < r^{-}_{\rm erg}$. The negative solution $r^{(-)}_{\rm erg}$ delimits the ergoregion in the region of negative radii, which spreads between the radii $r^{-}_{c} \leq r \leq r^{(-)}_{\rm erg} < 0$ in the equatorial plane, where r^{-}_{c} denotes the secondary cosmological horizon in the region $r < 0$. For more details on the properties of the ergosphere and its extension beyond the equatorial plane at $r < 0$, see [77]. In the region of positive radii $r > 0$, the inner radius $r^{+}_{\rm erg}$ delimits the boundary of the inner ergosphere, connected to the ring singularity which extends between radii $0 < r^{+}_{\rm erg}$, while the outer radius $r^{-}_{\rm erg}$ defines the boundary of the cosmological ergosphere, extending between the radii $r^{-}_{\rm erg} < r_c$. These radii are located at

$$r^{(-)}_{\rm erg} = -2\sqrt{\frac{1-a^2y}{3y}}\cos\left[\frac{1}{3}\arccos\sqrt{\frac{27y}{(1-a^2y)^3}}\right] \tag{20}$$

and

$$r^{\pm}_{\rm erg}(a^2, y) \equiv 2\sqrt{\frac{1-a^2y}{3y}}\cos\left[\frac{\pi}{3} \pm \frac{1}{3}\arccos\sqrt{\frac{27y}{(1-a^2y)^3}}\right]. \tag{21}$$

The parameters a^2, y of the spacetimes in which the two distinct radii $r^{\pm}_{\rm erg}$ exist are, according to Eq. (21), related by the condition

$$\frac{27y}{(1-a^2y)^3} \leq 1. \tag{22}$$

The equality in the relation (22) defines the function

$$y_{erg-s}(a^2) \equiv \frac{1}{a^2} + \frac{3\times 2^{1/3}}{2a^3}\frac{2^{1/3}(\sqrt{4+a^2}-a)^{2/3}-2}{(\sqrt{4+a^2}-a)^{1/3}}, \tag{23}$$

which in the $(a^2$-$y)$ parameter plane separates spacetimes with two separate ergospheres, the inner and the cosmological one, from spacetimes with a single one ergosphere extending from the ring singularity to the cosmological horizon. The spacetimes with a single or two separate ergoregions correspond to the region above/below the $y = y_{erg-s}(a^2)$ curve in the $(a^2$-$y)$ parameter plane. The function $y_{erg-s}(a^2)$ is also related to the existence of the static radius (see subsection below), which explains the label used.

Latitudinal coordinates $\theta_{\rm erg}$ of the ergosurface outside the equatorial plane depend on the radial coordinate according to the equation

$$\theta_{\rm erg}(r) = \arccos\sqrt{\frac{a^2y - 1 + \sqrt{I^2 - 4y\Delta_r}}{2a^2y}}. \tag{24}$$

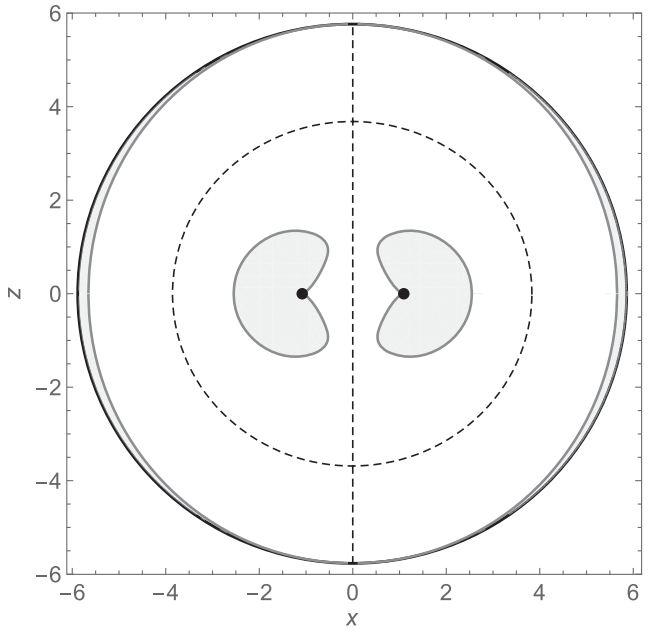


FIG. 2. Distribution of the ergosphere in KdS spacetime of class IVa with parameters $a^2 = 1.2, y = 0.02$, represented in the Kerr-Schild coordinates x,y,z. The figure shows a section through the plane y = 0 (here y not to be confused with the cosmological parameter y). The ergosphere is depicted as a gray area. The dashed vertical line depicts the spin axis, the black dots represent the incision through the ring singularity. The dashed ellipse depicts the static radius. The inner toruslike-shaped ergosphere connected to the ring singulrity can be seen, and the cosmological ergosphere, which spreads among the outer ergosurface and the cosmological horizon. The cosmological horizon is depicted as the outer black ellipse.

A detailed discussion of the properties of the ergosphere for relevant possible combinations of the KdS spacetime parameters a^2, y has been presented in [77] and will not be repeated here. We just recall that Eqs. (20), (21), and (24) hold assuming $a^2 y < 1$, which is the case of the class IV KdS spacetimes considered in this paper. The distribution of the ergosphere in the class IV KdS spacetimes is shown in Fig. 2; for other classes of the KdS spacetimes see [77].

D. Static radius

In this paper we intend to construct the shadows of superspinars observed by observers located at LNRFs at the so-called static radius r_s, where the gravitational attraction due to the central mass M is just compensated by the cosmological repulsion governed by the cosmological constant Λ. At and near the static radius the KdS spacetimes are closest to the asymptotically flat region of the Kerr spacetimes, and the LNRFs are close to the frames of static observers [82,83].

In dimensional units, the static radius depends on the mass and the cosmological constant as $r_s = (3GM^2/\Lambda c^2)$ [81], which can be simplified to the form

$$r_s = y^{-1/3} \tag{25}$$

in the dimensionless quantities we use. From Eq. (21) it follows that the static radius coincides with the radius where the two radii $r^{\pm}_{\mathrm{erg}}$ coalesce in the borderline case of spacetimes with $y = y_{erg-s}(a^2)$. Since at the static radius the only nonzero component of the freely falling observer's four-momentum P^μ is its time component P^t, while the other spatial components are zero, $P^i = 0$ [38], it follows that in the spacetimes with $y > y_{erg-s}(a^2)$ the static radius is limited outside the equatorial plane, since there is an ergosphere where necessarily $P^\phi > 0$.

III. EQUATIONS OF PHOTON MOTION

Since the components of the metric tensor $g_{\mu\nu}$ determined by the line element (2) are independent of time and azimuthal coordinates (t, ϕ), the conjugate momenta

$$k_t = g_{t\alpha} k^\alpha \equiv -E, \tag{26}$$

$$k_\phi = g_{\phi\alpha} k^\alpha \equiv = \Phi \tag{27}$$

of the photon four-momentum $k^\mu \equiv \mathrm{d}x^\mu/\mathrm{d}\lambda$, $x^\mu \in (t, r, \theta, \phi)$ are constants of motion. Here λ is the affine parameter and E, Φ are covariant "energy" and covariant "axial angular momentum"—in the pure Kerr case $y = 0$, where the spacetime is asymptotically flat, these are the energy and axial angular momentum as measured by distant static observers [87], while for $y > 0$ the spacetime is asymptotically de Sitter and these quantities are close to those measured by static observers at the static radius and its vicinity [81,82]. In addition to another motion constant, the rest mass m of the particle ($m = 0$ for photons), there is one more, the so-called fourth Carter constant $\mathcal{K}$, related to the hidden symmetry of the geometry, which must be nonnegative, $\mathcal{K} \geq 0$. The only case where $\mathcal{K} = 0$ occurs is for photons whose trajectories coincide with the spin axis $\theta = 0, \pi$ or for the so-called principal null congruence photons whose trajectories lie on the cones $\theta = \mathrm{const}$ [90].

As in previous works [28,68,77,81,91], for $E \neq 0$ it is convenient to introduce the modified motion constants X, q by the relations

$$X \equiv \ell - a, \tag{28}$$

where

$$\ell \equiv \Phi/E \tag{29}$$

is the impact parameter, and the modified fourth Carter constant

$$q \equiv \mathcal{K}/I^2E^2 - X^2, \tag{30}$$

which must vanish for the motion confined to the equatorial plane. Using the above motion constants, the Carter equations for photon motion become

$$\frac{\rho^2}{I^2E}k^t = \frac{(r^2+a^2)(r^2-aX)}{\Delta_r} + \frac{a(a\cos^2\theta + X)}{\Delta_\theta}, \tag{31}$$

$$\frac{\rho^2}{I^2E^2}k^r = \pm\sqrt{R}, \tag{32}$$

$$\frac{\rho^2}{I^2E^2}k^\theta = \pm\sqrt{W}, \tag{33}$$

$$\frac{\rho^2}{I^2E}k^\phi = \frac{a(r^2-aX)}{\Delta_r} + \frac{(a\cos^2\theta + X)}{\Delta_\theta \sin^2\theta}, \tag{34}$$

where

$$R(r) \equiv (r^2-aX)^2 - \Delta_r(X^2+q), \tag{35}$$

$$W(\theta) \equiv (X^2+q)\Delta_\theta - \frac{(a\cos^2\theta + X)^2}{\sin^2\theta}. \tag{36}$$

A detailed discussion of these equations of motion related to the both the KdS BH and NS spacetimes can be found in [28,77].

A. Allowed regions of motion constants

The photon motion is fully governed by the motion constants X, q, which in turn are related to the directional angles of a photon with respect to an emitter fixed to an observer (frame) and which govern the construction of the LEC related to the observer. Therefore, we have to determine for which values of these motion constants the motion of a photon released at a given position of the emitter, described by the pair (r_e, θ_e) of its radial and latitudinal coordinates, is actually allowed. At the starting point are the reality conditions

$$W(\theta_e; X, q) \geq 0, \tag{37}$$

$$R(r_e; X, q) \geq 0. \tag{38}$$

Using the linear dependence of the functions $R(r; X, q), W(\theta; X, q)$ on the motion constant q, similar to [68,77], we can rewrite the conditions (37) and (38) in the form

$$q_{\min}(X;\theta_e) \leq q \leq q_{\max}(X; r_e), \tag{39}$$

where

$$q_{\min}(X;\theta_e) \equiv \frac{\cot^2\theta_e}{\Delta_{\theta_e}}\left[(1-a^2y\sin^2\theta_e)X^2 + 2aX + a^2\cos^2\theta_e\right] \tag{40}$$

and

$$q_{\max}(X; r_e) \equiv \frac{(r_e^2 - aX)^2}{\Delta_{r_e}} - X^2. \tag{41}$$

It follows from Eq. (39) that the motion constants of a photon reaching the surface of the superspinar at $r = \mathcal{R}$ must satisfy the inequality

$$q \leq q_{\max}(X; \mathcal{R}). \tag{42}$$

We now briefly outline some properties of the functions $q_{\min}(X;\theta)$, $q_{\max}(X; r)$ for later reference.

1. Latitudinal function $q_{\min}(X;\theta_e)$

For the spacetime parameters satisfying $a^2y < 1$, the function $q_{\min}(X;\theta_e)$ is convex for $0 < \theta_e < \pi/2$, while for $\theta_e = \pi/2$ it holds $q_{\min}(X;\theta_e = \pi/2) \equiv 0$ [92]. The global minimum of the function $q_{\min}(X;\theta_e)$ is located at

$$X = X_{\min(\min)}(\theta_e) \equiv \frac{-a}{1-a^2y\sin^2\theta_e}, \tag{43}$$

and its value is

$$q = q_{\min(\min)}(\theta_e) \equiv \frac{-a^2\cos^2\theta_e}{1-a^2y\sin^2\theta_e}, \tag{44}$$

where $0 < \theta_e < \pi/2$. For $\theta_e \to 0$ the following relations hold: $X_{\min(\min)}(\theta_e) \to -a$, $q_{\min(\min)}(\theta_e) \to -a^2$, while for $\theta_e = 0$ (the motion along the spin axis) the reality condition $W(\theta; X, q) \geq 0$ reduces to

$$X = -a \quad (\ell = 0), \tag{45}$$

$$q \geq -a^2. \tag{46}$$

The behavior of the both functions $X_{\min(\min)}(\theta_e)$, $q_{\min(\min)}(\theta_e)$ is shown in Figs. 3 and 4. A detailed discussion of the properties of the function $q_{\min}(X;\theta)$ can be found, e.g., in [28,68,77].

2. Radial function $q_{\max}(X; r_e)$

In general, the function $q_{\max}(X; r_e)$ can be convex or concave depending on the sign of the coefficient $q_{\max(2)}$ of the quadratic term, where

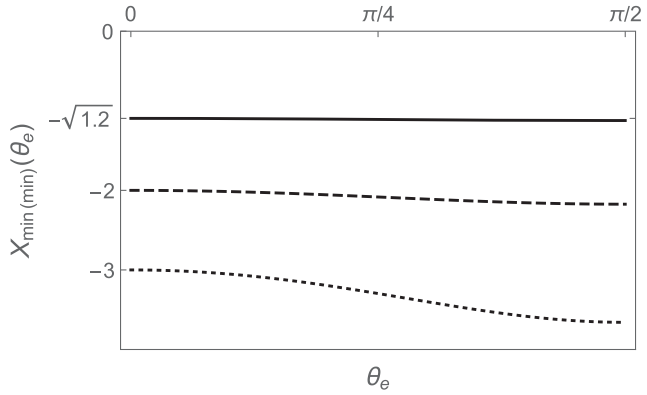


FIG. 3. Behavior of the function $X_{\min(\min)}(\theta_e)$ for fixed parameter $y = 0.02$ and varying parameter $a^2 = 1.2$ (full curve), $a^2 = 4$ (dashed curve), and $a^2 = 9$ (dotted curve).

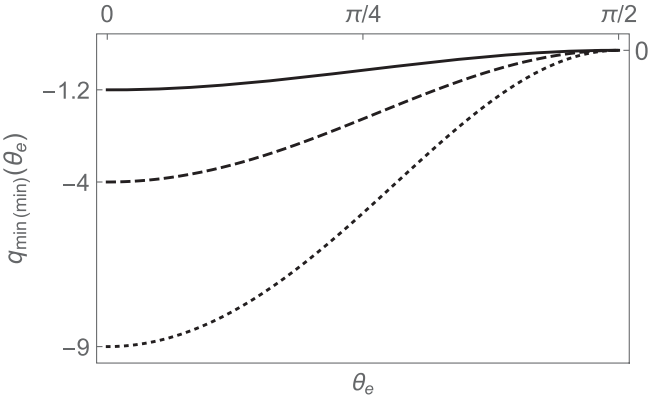


FIG. 4. Behavior of the function $q_{\min(\min)}(\theta_e)$ with the same styling and corresponding parameters as used in Fig. 3.

$$q_{\max(2)} = \frac{a^2 - \Delta_{r_e}}{\Delta_{r_e}}. \tag{47}$$

As follows from Eq. (47), at radii $r > 0$ that do (do not) interfere with the ergosphere, the function $q_{\max}(X; r_e)$ is convex (concave). In the first case, as shown in [28], the part of the plane of the (X-q) plane of motion constants described by the inequalities (39) and lying below the rising part of the function $q_{\max}(X; r_e)$ corresponds to locally counterrotating photons with positive impact parameter X and with negative energy, which are of course trapped by the gravitational potential. The parts of the (X-q) plane of motion constants described by the inequalities (39) and lying below the descending part of the $q_{\max}(X; r_e)$ function correspond to photons with $E > 0$ which can be trapped or escaping. In the latter case, the parts of the (X-q) plane of motion constants satisfying the inequalities (39) correspond only to photons with $E > 0$, which again can be escaping or trapped.

According to the relation (47), the function $q_{\max}(X; r_e)$ is concave at $r_{(erg)-} < r < 0$, and at $r = 0$ it is $q_{\max}(X; 0) = 0$.

B. Important cases of photon trajectories

The specific types of photon trajectories that have a major impact on the design of the resulting LECs have been discussed in detail in our previous work [77]. Here we briefly recall their properties. Some of the concepts introduced here are needed for the basic classification of the null geodesics in Sec. III C.

1. Spherical photon orbits

The spherical photon orbits (SPOs) are determined by the conditions

$$R(r) = 0, \qquad \frac{\mathrm{d}R}{\mathrm{d}r} = 0, \tag{48}$$

which, when solved simultaneously, give the motion constants for the photons on the SPOs

$$X = X_{\mathrm{SPO}}(r) \equiv \frac{r[(1 - ya^2)r^2 - 3r + 2a^2]}{a[2yr^3 + (ya^2 - 1)r + 1]}, \tag{49}$$

or, alternatively,

$$\ell = \ell_{\mathrm{SPO}}(r) \equiv \frac{Ir^3 - 3r^2 + a^2 Ir + a^2}{a[2yr^3 + (ya^2 - 1)r + 1]} \tag{50}$$

and

$$q = q_{\mathrm{SPO}}(r) \equiv -\frac{r^3}{a^2}\frac{y^2a^4r^3 + 2ya^2r^2(r+3) + r(r-3)^2 - 4a^2}{[2yr^3 + (ya^2 - 1)r + 1]^2}. \tag{51}$$

The zeros of the function $q_{\mathrm{SPO}}(r)$ give the radii of the equatorial circular photon orbits (ECPOs). It can be shown that in the case of the NS spacetimes there is only one ECPO at $r = r_{ph}^-$ in the region of positive radii, which is locally counterrotating in the LNRFs and unstable with respect to the radial perturbations. It gives the boundary of the region of the SPOs at $r > 0$. However, its expression is rather complex and therefore will not be presented here.

If $a^2 y < 1$, the functions $X_{\mathrm{SPO}}(r)$, $\ell_{\mathrm{SPO}}(r)$, $q_{\mathrm{SPO}}(r)$ have common points of divergence at radii $r_{d(SPO)}^{(-)}$, $r_{d(SPO)}^{\pm}$, where

$$r_{d(SPO)}^{(-)} = -2\sqrt{\frac{1 - a^2 y}{6y}}\cos\left[\frac{1}{3}\arccos\sqrt{\frac{27y}{2(1 - a^2 y)^3}}\right], \tag{52}$$

and

$$r_{d(SPO)}^{\pm} = 2\sqrt{\frac{1 - a^2 y}{6y}}\cos\left[\frac{\pi}{3} \pm \frac{1}{3}\arccos\sqrt{\frac{27y}{2(1 - a^2 y)^3}}\right], \tag{53}$$

for which the inequality $r^{(-)}_{d(SPO)} < 0 < r^{+}_{d(SPO)} < r^{-}_{d(SPO)}$ holds.

The radii $r^{\pm}_{d(SPO)}$ coalesce for $y = y_{d(SPO)}(y)$, where

$$y_{d(SPO)}(a^2) \equiv \frac{1}{a^2} - \frac{6}{\sqrt{2}a^3}\sinh\left[1/3\sinh^{-1}\frac{\sqrt{2}a}{2}\right]. \quad (54)$$

Note that the function $y_{d(SPO)}(y)$ in the $(y - a^2)$ plane determines the boundary between the KdS spacetime classes VI and VII, as introduced in [28]. For $y \lessgtr y_{d(SPO)}(y)$, the functions $X_{\rm SPO}(r)$, $q_{\rm SPO}(r)$ have two/no points of divergence [93].

The condition for unstable spherical orbits

$$\mathrm{d}^2R/\mathrm{d}r^2 < 0 \quad (55)$$

implies the inequality

$$8r(3yr^4 - I^2r^3 + 3(1 - a^2y)r^2 - 3r + a^2) > 0, \quad (56)$$

to be evaluated at the radii of the appropriate spherical orbit $r = r_{\rm SPO}$. Typical behavior of the functions $\ell_{\rm SPO}(r)$, $q_{\rm SPO}(r)$ for spacetime parameters corresponding to the class IV, and the related significant radii of the SPOs, are shown in Fig. 5.

The curve $(X_{\rm SPO}(r_{\rm SPO}), q_{\rm SPO}(r_{\rm SPO}))$ [hereafter $q_{\rm SPO}(X_{\rm SPO})$] parametrized by the radii $r_{\rm SPO}$ of the unstable spherical photon orbits plays a crucial role in the construction of LECs, since it gives the critical locus (cf., e.g., [94]) in the $X - q$ plane of the motion constants that determine the boundary between escaping and trapped photons.

The critical locus has essential importance in Sec. IV F, where we construct the KdS superspinar shadow for an observer at radius $r_o > r^{-}_{ph}$. This curve then corresponds to an arc or circle on the observer's celestial sky.

In the vicinity of stable SPOs there are bounded orbits where the photons, which we will call "trapped" (see Sec. III C), oscillate radially between two radial turning points. These orbits can be occupied if the source is located between these radii. Such a situation can occur if there is an overlap of the SPOs with the equatorial circular orbits of the test particles, allowing the possible occurrence of a thin Keplerian disk. In such a case the construction of LECs is important for the study of special optical effects such as self-illumination and self-eclipse or light echo [29].

2. Polar spherical photon orbits

The polar spherical photon orbits are the special case of the SPOs orbiting over the poles, crossing the symmetry axis. They are determined by the condition

$$\ell_{\rm SPO} = 0, \quad (57)$$

which can be expressed by the relation

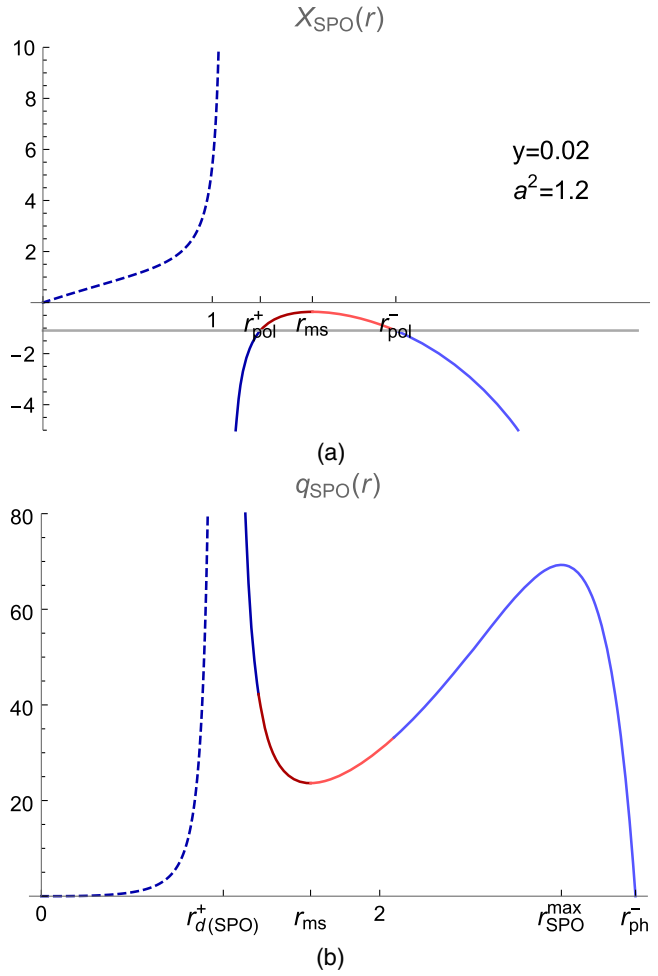


FIG. 5. Typical behavior of the functions $X_{\rm SPO}(r)$ and $q_{\rm SPO}(r)$ for parameters corresponding to class IV spacetime. The solid horizontal gray line in (a) indicates the value of $X = -a$ ($\ell = 0$). The dark blue dashed parts of these curves determine the motion constants of photons with negative covariant energy $E < 0$, delimited by the point of divergence $r^{+}_{d(SPO)}$ of both functions. The red/blue parts correspond to locally corotating/counterrotating SPOs, delimited by the radii $r^{\pm}_{\rm pol}$ of the polar SPOs, which are the zeros of the function $\ell_{\rm SPO}$. The dark/bright colors correspond to the stable/unstable SPOs that are separated by the radii r_{ms} of the marginally stable SPOs, which are the loci of the local maximum/minimum of the function $\ell_{\rm SPO}(r)/q_{\rm SPO}(r)$. The region of the SPOs is delimited by the radii of the equatorial circular photon orbit r^{-}_{ph}, which is a zero of the function $q_{\rm SPO}(r)$ (for more details see [77]).

$$y = y_{\rm pol}(r; a^2) \equiv \frac{r^2(3 - r) - a^2(1 + r)}{ra^2(r^2 + a^2)}. \quad (58)$$

The behavior of the function $y_{\rm pol}(r; a^2)$ is shown in Fig. 6 in comparison with the behavior of the function $y_h(r; a^2)$ for some appropriately selected values of the spin a^2. For given parameters a_0^2, y_0, the intersection of the line $y = y_0$ with the curve $y = y_{\rm pol}(r; a_0^2)$ gives the radii $r^{\pm}_{\rm pol}$ of the

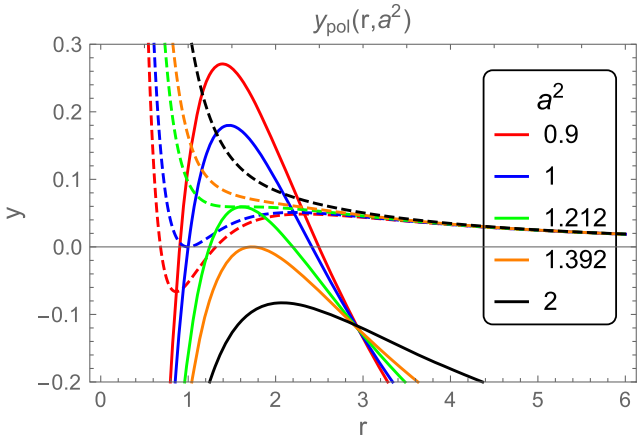


FIG. 6. Behavior of the function $y_{\rm pol}(r;a^2)$ (full curves) compared to the function $y_h(r;a^2)$ (dashed curves) for some representative values of the cosmological parameter a^2. For a given parameter a^2, both curves have the same color. It can be seen that the graph of the function $y_{\rm pol}(r;a^2)$ intersects the graph of function $y_h(r;a^2)$ at its local extrema, if any.

polar SPOs, which, as we have found in [77], reads

$$r^{\pm}_{\rm pol}=\frac{1}{I}\left[1+2\sqrt{1-\frac{a^2I^2}{3}}\cos\left(\frac{\pi\pm\psi}{3}\right)\right], \tag{59}$$

where

$$\cos\psi=\frac{a^2I^2-1}{\left(1-\frac{a^2I^2}{3}\right)^{3/2}}. \tag{60}$$

Figure 7 shows that there are KdS superspinar (NS) spacetimes with two polar SPOs, an inner stable one and an outer unstable one, which is the case of the superspinar (NS) spacetimes of classes IVa and VIa, introduced in [28], BH spacetimes with one unstable SPO, which corresponds to classes I–III, and superspinar (NS) spacetimes without polar SPOs, which correspond to classes VII and VIII. Figure 8 shows the limiting case of the superspinar (NS) spacetime, separating the spacetimes of classes IVa and VIa from the superspinar (NS) spacetimes with no polar SPO that correspond to classes IVb, V, and VIb. In this limiting case, the two polar SPOs merge. The radius $r_{\max(pol)}$ of this single polar SPO is given by the local maximum of the function $y_{\rm pol}(r;a^2)$, which, as follows from the analysis of Eqs. (59) and (60), increases linearly with the spin a as

$$r_{\max(pol)}(a)=a\sqrt{1+2/\sqrt{3}}. \tag{61}$$

The minimum value for $a=a_{crit(KdS)}$, $y=y_{crit(KdS)}$ is the same as the merging event horizons (see the green curves in Fig. 6), i.e.,

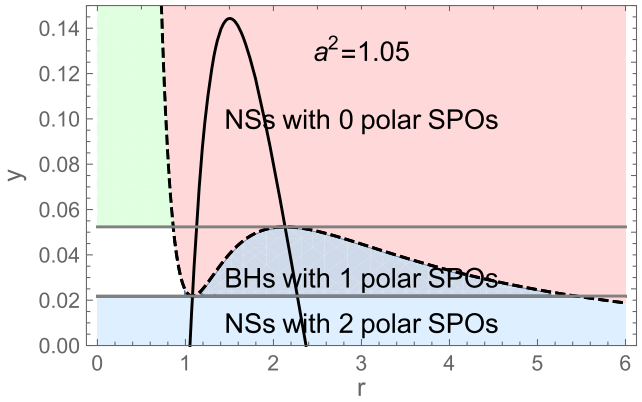


FIG. 7. Behavior of the function $y_{\rm pol}(r;a^2)$ (black full curve) for the spin parameter corresponding to the case $a^2_{SdS}<a^2<a^2_{\rm crit}$. The horizontal gray lines represent the values $y_{\min(h)}(a^2)$, $y_{\max(h)}(a^2)$ of the local minimum and maximum, respectively, of the function $y_h(r;a^2)$ (black dashed curve). For $0<y<y_{\min(h)}(a^2)$ (blue region), the function $y_{\rm pol}(r;a^2)$ determines two polar SPOs (the inner one stable, the outer one unstable) in the NS spacetimes belonging to class IVa or VIa. For $y_{\min(h)}<y<y_{\max(h)}(a^2)$ it determines one polar SPO (unstable) in the stationary region (gray area) of the BH spacetimes of class II or III. For $y_{\max(h)}(a^2)<y$ the curve $y=y_{\rm pol}(r;a^2)$ is irrelevant, since this part lies completely in the dynamic region (red area) of the NS spacetimes of class VII or VIII. The white/green area corresponds to the stationary region below the inner (Cauchy)/cosmological horizon of the appropriate BH/NS spacetime.

$$r_{\max(pol)}(a_{crit(KdS)})=r_i=r_o=r_c =\frac{1}{4}(3+2\sqrt{3})=1.616, \tag{62}$$

while the maximum value occurs for $a=a_{\max(pol)K}\equiv\sqrt{6\sqrt{3}-9}=1.180$, $y=0$, and reads

$$r_{\max(pol)}(a_{\max(pol)K})=1+\sqrt{4-2\sqrt{3}}=1.732, \tag{63}$$

where $a_{\max(pol)K}$ denotes the maximum spin that allows the existence of the polar SPOs, corresponding to the Kerr spacetime ($y=0$) (cf. [29]). Its square is $a^2_{\max(pol)K}=6\sqrt{3}-9=1.3923$. The parameters of the superspinar (NS) spacetimes with the single polar SPO at radii $r_{\max(pol)}$ are tied by the condition

$$\cos\psi=1 \tag{64}$$

in (60), which can be written in the form

$$y_{\max(pol)}(a^2)\equiv a^{-2}\left(\sqrt{\frac{a^2_{\max(pol)K}}{a^2}}-1\right). \tag{65}$$

The curve $y_{\max(pol)}(a^2)$ defines a boundary in the $(y\text{-}a^2)$ plane between the spacetimes of classes IVa and IVb,

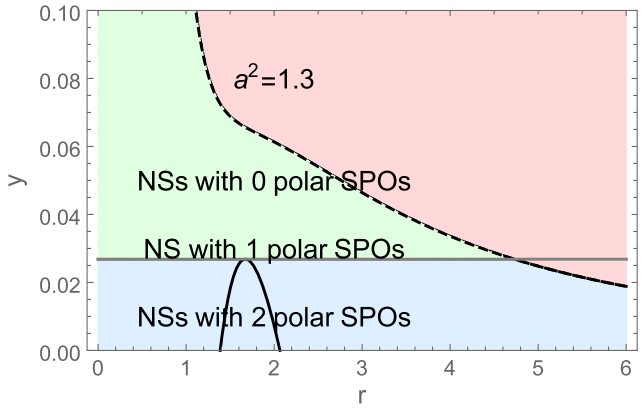


FIG. 8. Behavior of the function $y_{\mathrm{pol}}(r;a^2)$ (full curve) for the spin parameter corresponding to the case $a^2_{\mathrm{crit}} < a^2 < a^2_{\mathrm{max}(pol)K}$. The horizontal gray line represents the value $y_{\mathrm{max}(pol)}(a^2)$ of the local maximum of the function $y_{\mathrm{pol}}(r;a^2)$. The loci of this maximum represent the radius $r_{\mathrm{max}(pol)}(y)$ of the marginally stable polar SPO. This line separates the spacetimes with two polar SPOs, which belong to the class IVa or VIa, from the spacetimes with no polar SPOs, which correspond to the class IVb or VIb. For $a^2_{\mathrm{max}(pol)K} < a^2$, $y_{\mathrm{max}(pol)}(a^2) < 0$, so there are no polar SPOs in the spacetimes with $a^2_{\mathrm{max}(pol)K} < a^2$. The coloring has the same meaning as in the previous figure. The division of the parameter plane $(y - a^2)$ into regions corresponding to the different types of KdS spacetimes discussed above is shown in Fig. 9.

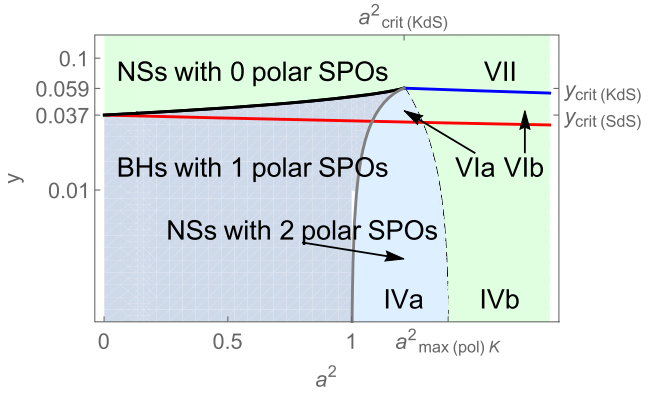


FIG. 9. Functions $y_{\mathrm{max}(h)}(a^2)$ (black curve), $y_{\mathrm{min}(h)}(a^2)$ (gray curve), $y_{\mathrm{max}(pol)}(a^2)$ (black dashed curve), $y_{erg-s}(a^2)$ (red curve), $y_{d(SPO)}(a^2)$ (blue curve) and the corresponding division of the parameter plane $(y\text{-}a^2)$ into sections corresponding to BH and NS spacetimes with appropriate number of polar SPOs. The coloring of each region is chosen to be consistent with Figs. 7 and 8. The red and blue curves represent the subdivision of the parameter plane according to the classification introduced in [28], which is recalled below and in the commentary to Fig. 10. Note that the values on the y axis are plotted on a logarithmic scale.

which correspond to the superspinar (NS) spacetimes with two polar SPOs, and the spacetimes of classes VIa and VIb, corresponding to the superspinar (NS) spacetimes with no polar SPOs, as introduced in [28].

In this paper, we do not address all the properties that trigger this classification, except for the presence/absence of polar SPOs. In Fig. 9 we therefore present the classification of spacetimes only in terms of the number of horizons and polar SPOs.

C. Classification of the KdS superspinar (NS) spacetimes

The classification of KdS superspinar (NS) spacetimes into classes IV–VIII, some of which are indicated in Fig. 9, is determined by the nature of the radial motion of the photons. This motion is determined by the behavior of the effective potentials $X_{\pm}(r;q,y,a^2)$, where the equality

$$X = X_{\pm}(r;q,y,a^2) \equiv \frac{ar^2 \pm \sqrt{\Delta_r[r^4 + q(a^2-\Delta_r)]}}{a^2-\Delta_r} \quad (66)$$

gives the turning points in the radial direction for photons with the impact parameter X. Equation (66) follows from the inequality (38) by inserting an equal sign and expressing the parameter X.

The curves $X = X_{\pm}(r;q,y,a^2)$ delimit the region of forbidden values of the impact parameter X in the $(r\text{-}X)$ plane, where the inequality (38) is not met, acting as a repulsive potential barrier of the radial photon motion. It follows from the definition in Eq. (66) and from the discussion made in Sec. II C that the function $X_{+}(r;q,y,a^2)$ can have up to two divergent points in the region $r > 0$ [95] that coincide with the solution of Eq. (19), i.e., with radii given by Eq. (21). Note that these radii depend only on the spacetime parameters a^2, y. If these two real solutions exist, we have the KdS spacetimes with the divergent repulsive barrier (DRB) of the radial photon motion. Thus, it further follows from Sec. II C that spacetimes with DRB correspond to spacetimes with two separate ergospheres—the inner one and the cosmological one. On the other hand, if there is no real solution of Eq. (19), the function $X_{+}(r;q,y,a^2)$ has no points of divergence and we have the KdS spacetimes with the restricted repulsive barrier (RRB) of the radial photon motion. It therefore corresponds to the spacetimes with a single ergosphere. The limiting cases between the KdS spacetimes with DRB and RRB, where the radii given by Eq. (21) merge, have spacetime parameters restricted by the relation $y = y_{erg-s}(a^2)$, which is represented by the red curve in Fig. 9. The spacetimes with DRB/RRB correspond to the region below/above the curve $y = y_{erg-s}(a^2)$ in the $(a^2\text{-}y)$ parameter plane.

A detailed analysis of the $X_{\pm}(r;q,y,a^2)$ functions has been carried out in [28] for $r > 0$ and will not be repeated here. Using the typical behavior of the potentials shown in Fig. 10, we briefly outline here the basic characteristics of

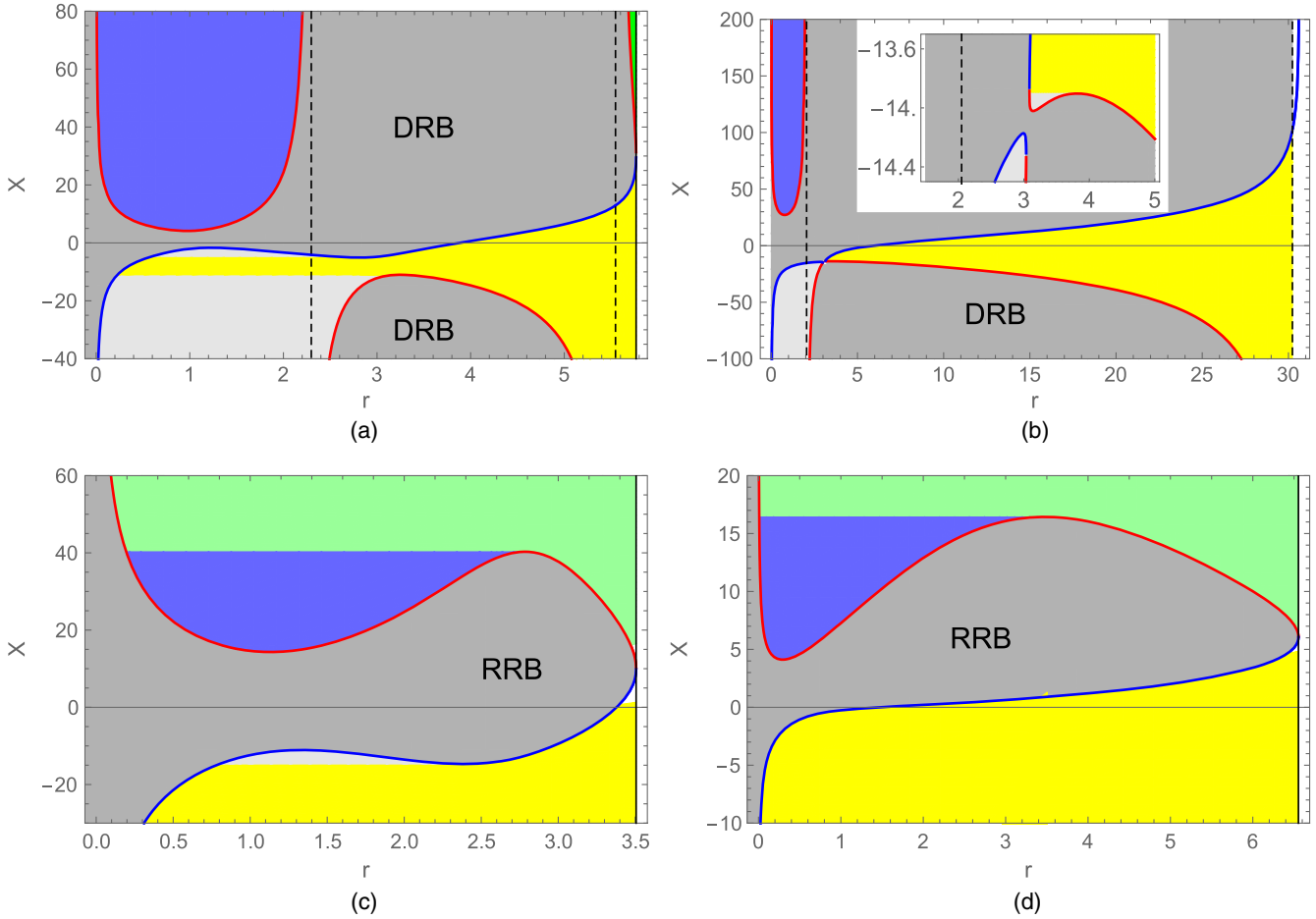


FIG. 10. Basic types of photon orbits in individual classes of the KdS superspinar (NS) spacetimes with appropriately chosen parameters y, a^2 in relation to the effective potentials $X_+(r; q, y, a^2)$ (red curve) and $X_-(r; q, y, a^2)$ (blue curve). The motion constant q is chosen to highlight the different types of orbits. The curves $X_\pm$ delimit the region of forbidden values of the impact parameter X (dark gray region), acting as a barrier of radial motion, which can be divergent (DRB) or restricted (RRB). The blue region corresponds to bound orbits occupied by trapped photons with negative covariant energy $E < 0$ (see [28] for details), the green region corresponds to unbound orbits occupied by photons with $E < 0$ but escaping toward the cosmological horizon. Light gray regions mark the trapped photons with $E > 0$, yellow regions mark the escaping photons with at most one turning point, if initially directed inward. Black full vertical lines at the common points of the functions $X_\pm(r)$ mark the loci of the cosmological horizon. Black dashed vertical lines are the asymptotes of the function $X_+(r)$ at $r = r^\pm_{\rm erg}$, delimiting the radial extent of the ergosurface in the equatorial plane. Only class IV is relevant from an astrophysics perspective, since the other classes correspond to enormously high values of a cosmological parameter that is inconsistent with observations and will therefore not be discussed in detail. The range and structure of the photon orbits in the case of class IVa are as follows. The local minima of the function $X_+(r)$ correspond to the stable spherical photon orbits with $E < 0$, which appear to be retrograde in the LNRFs. Varying the motion constant q, these minima correspond to the blue dashed part of the curve $X_{\rm SPO}(r)$ in Fig. 5(a). Their radii $r^{E<0}_{SPOs}$ are in the range $0 < r^{E<0}_{SPOs} < r^+_{d(SPO)}$. The local maxima of the function $X_-(r)$ correspond to the stable SPOs with $E > 0$, which can be either retrograde or prograde in the LNRFs, with the polar spherical orbit at $r = r^+_{\rm pol}$ as being the boundary between these directions. Varying the motion constant q, these local maxima correspond to the dark blue and red parts of the curve $X_{\rm SPO}(r)$ in Fig. 5(a). The radii $r^{E>0}_{SPOs}$ are in the range $r^+_{d(SPO)} < r^{E>0}_{SPOs} < r_{ms}$. The local maxima of the function $X_+(r)$ and the local minima of the function $X_-(r)$ represent the unstable SPOs. With varying motion constant q, these extremes correspond to the light blue and red parts of the $X_{\rm SPO}(r)$ curve in Fig. 5(a). These extrema meet each other at $r = r^{\rm max}_{\rm SPO}$, where $r^{\rm max}_{\rm SPO}$ is the local maximum of the function $q_{\rm SPO}(r)$. Varying the motion constant q, the projection of these maxima onto the (q-X) motion constant plane forms the critical locus. The radii $r_{\rm SPO}$ of the unstable SPOs are in the range $r_{ms} < r_{\rm SPO} < r^-_{ph}$, where r^-_{ph} is the radii of the equatorial circular photon orbit, which is counterrotating. The structure of the photon orbits in other cases can be found, e.g., in [28,77]. (a) IVa: $y = 0.02$, $a^2 = 1.2$; $q = 60$. (b) V: $y = 0.001$, $a^2 = 20$. (c) VIb: $y = 0.04$, $a^2 = 1.5$; $q = 500$. (c) VII: $y = 0.02$, $a^2 = 50$; $q = 0.1$.

the individual KdS superspinar (NS) spacetimes for the following suitably chosen representatives:

(i) Class IVa, represented by the spacetime with parameters $y = 0.02$, $a^2 = 1.2$ [see Fig. 10(a)], corresponding to the spacetimes with DRB and the regions of trapped photon orbits with both positive and negative covariant energy (see the comment on Fig. 10). The label "a" marks the presence of two polar SPOs, which is manifested by the fact that local extrema $X_{ex(-)}$ of the function $X_{-}(r; q, y, a^2)$ can take the value $X_{ex(-)} = X_{\rm SPO} = -a$. The behavior of the effective potentials corresponding to Class IVb, not shown in Fig. 10, where the label "b" corresponds to the absence of the polar SPOs, is qualitatively the same as for class IVa. The absence of the polar SPOs is manifested by the extrema $X_{ex(-)} < -a$.

(ii) Class V, represented by the spacetime with parameters $y = 0.001$, $a^2 = 20$ [see Fig. 10(b)], corresponding to the spacetimes with the same properties of the radial photon motion as in the previous case, but besides that it contains a small region of unusual bound orbits with $X_{+} < X < 0$ and $E > 0$ (see detail in the cutout)—for $r_e < r^{\rm max}_{\rm SPO}$ the critical locus $X_{\rm SPO}(q_{\rm SPO})$ is partly hidden in the forbidden region.

(iii) Class VIb, represented by the spacetime with parameters $y = 0.04$, $a^2 = 1.5$ [see Fig. 10(c)], corresponding to the spacetimes with RRB of the radial photon motion and the regions of trapped photon orbits with both positive and negative covariant energy. The label b again corresponds to the absence of the polar SPOs. The behavior of the effective potentials for the both classes VIa and VIb differs in the same manner as in the cases of IVa and IVb.

(iv) Class VII, represented by the spacetime with parameters $y = 0.02$, $a^2 = 50$ [see Fig. 10(d)], corresponding to the spacetimes with no trapped photon orbits with positive covariant energy. The other properties of the photon motion are the same as for class VI.

Class VIII corresponds to spacetimes with extremely high spacetime parameters, for which $a^2 y > 1$. It is not shown in Fig. 10, since the behavior of the effective potentials is qualitatively the same as for class VII. However, it differs from the class VII by the properties of the latitudinal photon motion (for more details see [28]).

In the present paper we do not construct LECs for the KdS spacetime classes VI–VIII because of the extremely outlying values of the spacetime parameters relative to the parameters corresponding to cosmological observations of the relict cosmological constant [96], for which any connection with real observations is excluded.

IV. LIGHT ESCAPE CONES AND KDS SUPERSPINAR SHADOWS IN THE LNRFS

In this section, we present the resulting LECs/shadows for some selected KdS superspinar (NS) spacetimes that belong to classes IV and V, with astrophysically acceptable values of the cosmological parameter. However, in order to highlight the optical effects, we use a cosmological parameter many orders of magnitude above the realistic values, which qualitatively does not change the result. Emitters' positions are chosen to give qualitatively different resulting LECs.

The shape of the LEC for a given spacetime depends on the latitudinal and radial position of the emitter. We present the construction of the LEC for the special latitudinal positions, namely, on the spin axis ($\theta_e = 0°$), in the equatorial plane ($\theta_e = 90°$), and we choose characteristic latitudinal coordinates of $\theta_e = 60°$ or $\theta_e = 70°$ representing a general position of the observer, depending on the better representation of the shadow structure. In order to clearly display a continuous progression of the shape of the LEC as the latitude increases, some specific latitudes are used in addition to those mentioned above.

The LECs are constructed in greatest detail for accordingly chosen radii in the case of the spacetimes of class IVa, which has the most complex structure of null geodesics according to the radial photon motion. The outstanding radii, which give some qualitative changes of the LECs, are presented in Tables I–III.

TABLE I. Significant radii related to the photon motion in the KdS spacetime of class IVa.

y	a^2	$r^{+}_{d(SPO)}$	$r^{+}_{\rm pol}$	r_{ms}	$r^{-}_{\rm pol}$	$r^{+}_{\rm erg}$	r^{-}_{ph}	r_s	$r^{-}_{\rm erg}$	r_c
0.02	1.2	1.08	1.28	1.59	2.08	2.30	3.51	3.68	5.55	5.77

TABLE II. Significant radii related to the photon motion in the KdS spacetime of class IVb.

y	a^2	$r^{+}_{d(SPO)}$	r_{ms}	$r^{+}_{\rm erg}$	r^{-}_{ph}	r_s	$r^{-}_{\rm erg}$	r_c
0.02	2	1.10	1.93	2.36	3.58	3.68	5.44	5.81

TABLE III. Significant radii related to the photon motion in the KdS spacetime of class V.

y	a^2	$r^{+}_{d(SPO)}$	$r^{+}_{\rm erg}$	$r^{\rm max}_{\rm SPO}$	r_{ms}	r^{-}_{ph}	r_s	$r^{-}_{\rm erg}$	r_c
0.001	20	1.02	2.05	3.06	3.57	6.24	10	30.23	30.59

A. Directional angles of photons in LNRFs

Using the locally measured photon four-momentum components $k^{(a)}$, the standard (α, β, γ) directional angles can be introduced using the relations [30]

$$\cos\alpha = k^{(r)}/k^{(t)}, \tag{67}$$

$$\sin\alpha\cos\beta = k^{(\theta)}/k^{(t)}, \tag{68}$$

$$\sin\alpha\sin\beta = \cos\gamma = k^{(\phi)}/k^{(t)}. \tag{69}$$

The frame and the coordinate components are related by the general formula

$$k^{(a)} = \omega^{(a)}_{\mu} k^{\mu}, \tag{70}$$

where the differential 1-forms of the LNRFs are given by relations

$$\omega^{(t)} = \sqrt{\frac{\Delta_r \Delta_\theta \rho^2}{I^2 A}}\mathrm{d}t, \tag{71}$$

$$\omega^{(r)} = \sqrt{\frac{\rho^2}{\Delta_r}}\mathrm{d}r, \tag{72}$$

$$\omega^{(\theta)} = \sqrt{\frac{\rho^2}{\Delta_\theta}}\mathrm{d}\theta, \tag{73}$$

$$\omega^{(\phi)} = \sqrt{\frac{A\sin^2\theta}{I^2\rho^2}}(\mathrm{d}\phi - \Omega_{LNRF}\mathrm{d}t) \tag{74}$$

and the coordinate four-momentum components $k^{\mu} = \frac{\mathrm{d}x^{\mu}}{\mathrm{d}\lambda}$, $x^{\mu} \in \{t, r, \theta, \phi\}$ are given by Eqs. (31)–(36).

For completeness, we present the orthonormal tetrad of basis vectors,

$$e_{(t)} = \sqrt{\frac{I^2 A}{\Delta_r \Delta_\theta \rho^2}}\left(\frac{\partial}{\partial t} + \Omega_{LNRF}\frac{\partial}{\partial \phi}\right), \tag{75}$$

$$e_{(r)} = \sqrt{\frac{\Delta_r}{\rho^2}}\frac{\partial}{\partial r}, \tag{76}$$

$$e_{(\theta)} = \sqrt{\frac{\Delta_\theta}{\rho^2}}\frac{\partial}{\partial \theta}, \tag{77}$$

$$e_{(\phi)} = \sqrt{\frac{I^2\rho^2}{A\sin^2\theta}}\frac{\partial}{\partial \phi}. \tag{78}$$

Here Ω_{LNRF} is the angular velocity of the LNRF relative to distant static observers, which is given by the relation

$$\Omega_{LNRF} = -\frac{g_{t\phi}}{g_{\phi\phi}} = \frac{a[\Delta_\theta(r^2+a^2) - \Delta_r]}{A}. \tag{79}$$

Through Eqs. (67)–(74) and the associated relations, there is a correspondence between the triad of the directional angles (α, β, γ) and the double of the motion constants (X, q). In fact, any two directional angles from this triad are sufficient; the standard choice is the pair (α, β). The construction of the LECs can thus be regarded as a transformation of the $(X$-$q)$ plane of the motion constants onto the directional angles [68]. To give an insight into the notion of the motion constants, we present the corresponding $(X$-$q)$ plane together with each constructed LEC.

B. Representation of the LECs

In the present paper, we represent the LECs as two-spheres represented in 3D, following the work [68], where we were more concerned by their performance in 2D mode. These spheres consist of the end points of the photon's three-momentum vector of unit magnitude $k^{(t)} = 1$ located at its center, where the emitter resides; the directional angles are given by the relations (67)–(69) corresponding to all the motion constants (X, q) allowing the motion at the considered emitter's location. This embodiment allows us to show more details of a given situation without distortion, but the disadvantage is that we need two views for each situation so that no hidden parts of the spheres remain. Therefore, in the figures below we present for each situation the plane of motion constant $(X$-$q)$ and the corresponding two-sphere shown from two different views.

C. Complementary cones to the LECs

When constructing the LECs, we can distinguish the photons that escape toward the cosmological horizon from those that do not, thus forming a complementary cone to the LEC. While the concept of the LEC is clear for both black holes and superspinars (NSs), in the case of complementary cones there are differences between the BH, superspinar, and NS spacetimes.

In the case of BH spacetimes the situation is much simpler due to the presence of the BH event horizon, which acts as a one-way membrane, and due to the absence of stable SPOs and related bound photon orbits—the photon trajectories in BH spacetimes can have at most one radial turning point. The total light flux from a light emitter consists of photons escaping to the cosmological horizon, either directly or after passing through some pericenter, forming thus the LEC, and photons absorbed by the black hole, which again reach the event horizon either directly or after passing through some apocenter; these photons form the complementary cone to the LEC, i.e., the captured cone. The boundary between these cones is given by the photons emitted toward the unstable SPOs.

1. Classification of the complementary cones in superspinar (NS) spacetimes

In the case of the superspinar (NS) spacetimes, the notion of the LECs is the same as in the case of BH spacetimes. However, the cone complementary to the LEC has a more complex character even in the case of KdS superspinars, where the surface is assumed to serve as a one-way membrane, similar to the BH spacetimes, because of the existence of stable SPOs. The case of the KdS NSs is even more complex, as there is no surface serving as a one-way membrane. We propose the following classification of the complementary cones to the LECs in the case of superspinars and NSs.

KdS superspinars:

(i) "trapped cones," formed by bound trajectories of photons that oscillate between two radial turning points r_1, r_2, $0 < r_1 < r_2 < +\infty$, i.e., the pericenter and apocenter, around some stable SPO; in this case, the light source must be in a region of stable SPOs; and

(ii) "engrossed cones," formed by trajectories terminating on the superspinar surface.

KdS NS spacetimes:

(i) "repelled escape cones," consisting of photons passing through the interior of the ring singularity at $r = 0, 0 \le \theta < \pi/2$ into the region of negative radii and, after passing through a certain radial turning point at $r_t < 0$, finally repelled back to $r > 0$; and

(ii) "oppositely escape cones" composed of photons passing through the interior of the ring singularity at $r = 0, 0 \le \theta < \pi/2$ into the region of negative radii and continuing to the other infinity $r \to -\infty$.

In the present article we will not discuss the last two classes (NS spacetime) of light cones in detail, as we have covered them in [77]. Here we assume that light does not reach the negative radius region, since the surface of the superspinar prevents it.

2. Negative energy light cones

It should be added that there is another criterion for classifying photon orbits according to the sign of their covariant energy E. As we have found in [28], in each KdS spacetime there exist orbits, where the photons have positive impact parameter X, but appear to move in retrograde direction as seen in the LNRFs. These orbits must be located in the ergosphere, and the photons must have negative covariant energy $E < 0$ [97]. In [28] we found that their impact parameter X must satisfy $X > X_+(r_e; q \ge 0)$. In spacetimes with DRB of the radial photon motion, i.e., in the case of classes IV and V, there are two spatially separated ergoregions (see Fig. 2). The inner ergoregion is occupied by the trapped negative energy photons, which we have shown in Figs. 10(a) and 10(b) as the blue part of the (X-r) plane, while the cosmological ergoregion (see Sec. II C) is occupied by the negative energy photons escaping toward the cosmological horizon, which we have depicted as the green part of the (X-r) plane in Figs. 10(c) and 10(d).

In spacetimes with RRB of the radial photon motion, i.e., in the case of classes VI–VIII, the ergoregion spreads along whole equatorial plane. For $r_e < r^-_{d(SPO)}$ there are photons with $E < 0$ which can be trapped or escape, for $r^-_{d(SPO)} < r_e < r_c$ there exist only escaping photons with $E < 0$ [see Figs. 10(c) and 10(d)].

D. Description of the light cones

We give the structure of the light cones for the KdS superspinars, and for comparison we also include the parts corresponding to the KdS NSs, where the photon can reach the region of negative radii.

The escape cones in Figs. 13–24 are shown as the yellow part of the two-sphere, which we can regard as the emitter's local celestial sphere, the trapped cones are shown in gray. The blue parts correspond to the trapped photons with negative covariant energy $E < 0$, which occurs when the source is in the inner ergosphere. The green parts depict the photons with negative covariant energy escaping toward the cosmological horizon when the source is in the outer ergosphere. The boundary between the trapped and escape light cones, given by a beam directed toward the unstable spherical photon orbits, i.e., with motion constants $X_{\rm SPO}$, $q_{\rm SPO}$, is illustrated by full or dashed orange curves. The dashed curve denotes that the photon has a radial turning point before it reaches the unstable SPO. The darker purple parts correspond to the escape cones to the other infinity; the lighter purple parts correspond to the repelled photons with radial turning points at negative radii $r < 0$. The engrossed cones are depicted in red; in Figs. 13–24 the surface radius of the superspinar is set to $\mathcal{R} = 0.1$.

Note that the coloring in Figs. 13–24 is chosen to match the coloring in the attached (X-q) planes and the corresponding parts in the ($X - r$) plots in Fig. 10.

E. Dependence of LEC's shape on the emitter's position

1. Latitudinal position on the spin axis

If the emitter is located on the spin axis, then all emitted photons have the impact parameter $X = -a$ ($\ell = 0$) and the motion constant q can take the values $-a^2 \le q \le q_{\rm max}(X = -a; r_e)$ [see Figs. 11(a), 13(a), 16(a), 17(a), 19(a), 22(a), and 23(a)]. The inward directed photons with $-a^2 \le q < 0$ enter the region of negative radii $r < 0$, from which the photons with $-a^2 \le q \le q_{\rm SPO}(r_{\rm SPO} < 0)$ escape to the other infinity ($r \to -\infty$), photons with $q_{\rm SPO}(r_{\rm SPO} < 0) < q < 0$ have the radial turning point at $r_t < 0$ and are repelled back to the $r > 0$ region, and photons with $q = 0$ hit the ring singularity. For $0 < q \le q_{\rm max}(X = -a; \mathcal{R})$, the radiation is engrossed by the

superspinar surface at $r = \mathcal{R}$. Other photons may escape or be trapped depending on the radial position with respect to the radii of the outer polar unstable SPO at $r^{-}_{\rm pol}$, if it exists.

In Figs. 11(a) and 12(a) the outstanding values of the motion constant q, i.e., $-a^2$, $q_{\rm SPO}(r_{\rm SPO} < 0)$, 0, $q_{\rm max}(X = -a; \mathcal{R})$, $q_{\rm max}(X = -a; r_e)$ are denoted as q_{-2}, q_{-1}, q_0, q_1, q_2, respectively.

If $r_e < r^{+}_{\rm pol}$ is small enough such that $q_{\rm max}(X = -a; r_e) < q_{\rm SPO}(X_{\rm SPO} = -a)$ [Fig. 11(a) left] or there are no polar SPOs [Fig. 23(a) left], then all photons with $q > 0$ escape [Figs. 12(a) and 23(a) right].

For larger values of r_e, but still satisfying the condition $r_e < r^{-}_{\rm pol}$, such that $q_{\rm max}(X = -a; r_e) > q_{\rm SPO}(X_{\rm SPO} = -a)$ [cf. black and orange points of the $(X - q)$ motion constant plane in Figs. 13(a) and 16(a)], there are trapped cones, which are depicted as a spherical belt. They are delimited by photons emitted directly outward to $r^{-}_{\rm pol}$, represented by full orange circles, and indirectly inward with radial turning point r_t, $r_t < r_e$, which are represented by the dashed orange circles [Figs. 13(a) and 16(a)]. The trapped cones then exist for any latitude.

If $r_e > r^{-}_{\rm pol}$, there are no trapped photons, but there are inward directed photons terminating on the outer polar SPO [see the orange circle in Figs. 17(a), 19(a), and 20(a)].

In the spacetimes without polar SPOs, all photons either escape or are engrossed by the superspinar surface, independent of the radial position of the emitter [see Fig. 23(a)].

2. General latitudinal position

Depending on the latitude of the emitter, the trapped cones can take the shape of a spherical belt [see Figs. 13 and 16(a)], biangle [Figs. 14(a), 16(b), and 16(c)], spherical canopy (Fig. 24), or cover substantial part of the emitter's sky on a hemisphere oriented in the direction opposite to that of the LNRF's motion [see Figs. 12(b), 14(b), 15, and 18]. The boundary of the cone is made up of two lines, one corresponding to photons with no radial turning point before reaching the unstable SPOs, which we present as a full orange curve, and the other corresponding to photons with one radial turning point, which we present as a dashed orange curve.

If the emitter is in the region of stable SPOs, i.e., under the radii r_{ms} in spacetimes of classes IV–VI, or under the radii $r^{\rm max}_{\rm SPO}$ in the case of classes VII and VIII (see the comment to Fig. 10), then all the unstable SPOs are located above the emitter, and for every outward photon that directly reaches an unstable SPO, there exists an inward photon with the same motion constants $X_{\rm SPO}$, $q_{\rm SPO}$ that reaches the same orbit after passage through some radial turning point. The two lines forming the boundary of the trapped cone are then placed symmetrically on the emitter's sky with respect to the $(\mathbf{e}_\theta$-$\mathbf{e}_\phi)$ plane of his local vector basis (see Figs. 13–15). The photon sent on the actual stable SPO, thus maintaining the actual radius r_e, is represented by a blue dot.

If the emitter is located in the region of unstable SPOs, i.e., at $r_{ms} < r_e < r^{-}_{ph}$ for class IVa, some unstable SPOs are located below the emitter's location and can only be reached directly by photons emitted in the radial inward direction, while the unstable SPOs located above can be reached directly by outwardly directed photons, or indirectly by inwardly directed photons, as described above. The full orange curve then exceeds the dashed one and forms an arc [see Figs. 16(c), 17(c), 18, and 20]. The trapped cone appears for latitudes exceeding some critical value $\theta_{{\rm max}(circ)}$ and has a shape of spherical biangle. The tips of the biangles, depicted as white dots, represent the directions of photons sent into the currently occupied unstable SPO.

For the location of the emitter above the region of unstable SPOs, i.e., at $r_e > r^{-}_{ph}$, only inwardly directed photons can reach the unstable SPOs, hence the dashed orange line and the trapped cone disappear. The full orange curve then forms a circle for $0 \leq \theta_e < \theta_{{\rm max}(circ)}$ or an arc for $\theta_{{\rm max}(circ)} \leq \theta_e \leq \pi/2$ about the engrossed cone (see Figs. 21 and 22).

We can emphasize that in the spacetimes with the polar SPOs there exist some maximum angle $\theta_{{\rm max}(circ)}$ for which the circular edges of the trapped cones making the biangles disconnect [Figs. 14(a), 16(a), 18, and 17(c)], or the circle around the engrossed cone splits into an arc [cf. Figs. 21(a) and 21(b)].

In the spacetime with no polar SPOs, there exists some minimum angle $\theta_{{\rm min}(arc)}$ which must be exceeded for occurrence of the arc [cf. Figs. 23(b) and 23(c)].

Since the LECs are closely related to the superspinar shadows, it is more interesting from an astrophysical point of view that the angle $\theta_{{\rm max}(circ)}$ can be regarded as a maximum angle for which an observer in a spacetime with the polar SPOs at $r_o > r^{-}_{ph}$ can see a distortion of the celestial background along a circular contour on his celestial sphere, which changes to an arc at larger latitudes. Observing the circle can thus give upper bounds either on the latitude of the observer or on the spin of the observed object for a known value of the second parameter.

Accordingly, the angle $\theta_{{\rm min}(arc)}$ can be regarded as the minimum angle at which an observer at $r_o > r^{-}_{ph}$ can see the arc in spacetimes without polar SPOs. Observing an arc with a small central angle ξ can give a lower limit on the spin parameter a^2 (see Sec. IV F).

Note that in all types of the KdS superspinar (NS) spacetimes there is a limiting angle $\theta_{{\rm max}(ell)}$ for which the shadow of a superspinar observed by a distant observer, i.e., at $r_e > r^{-}_{ph}$, has the shape of an ellipse (Fig. 21). At larger observer latitudes, the shadow gradually transforms into a wedge shape.

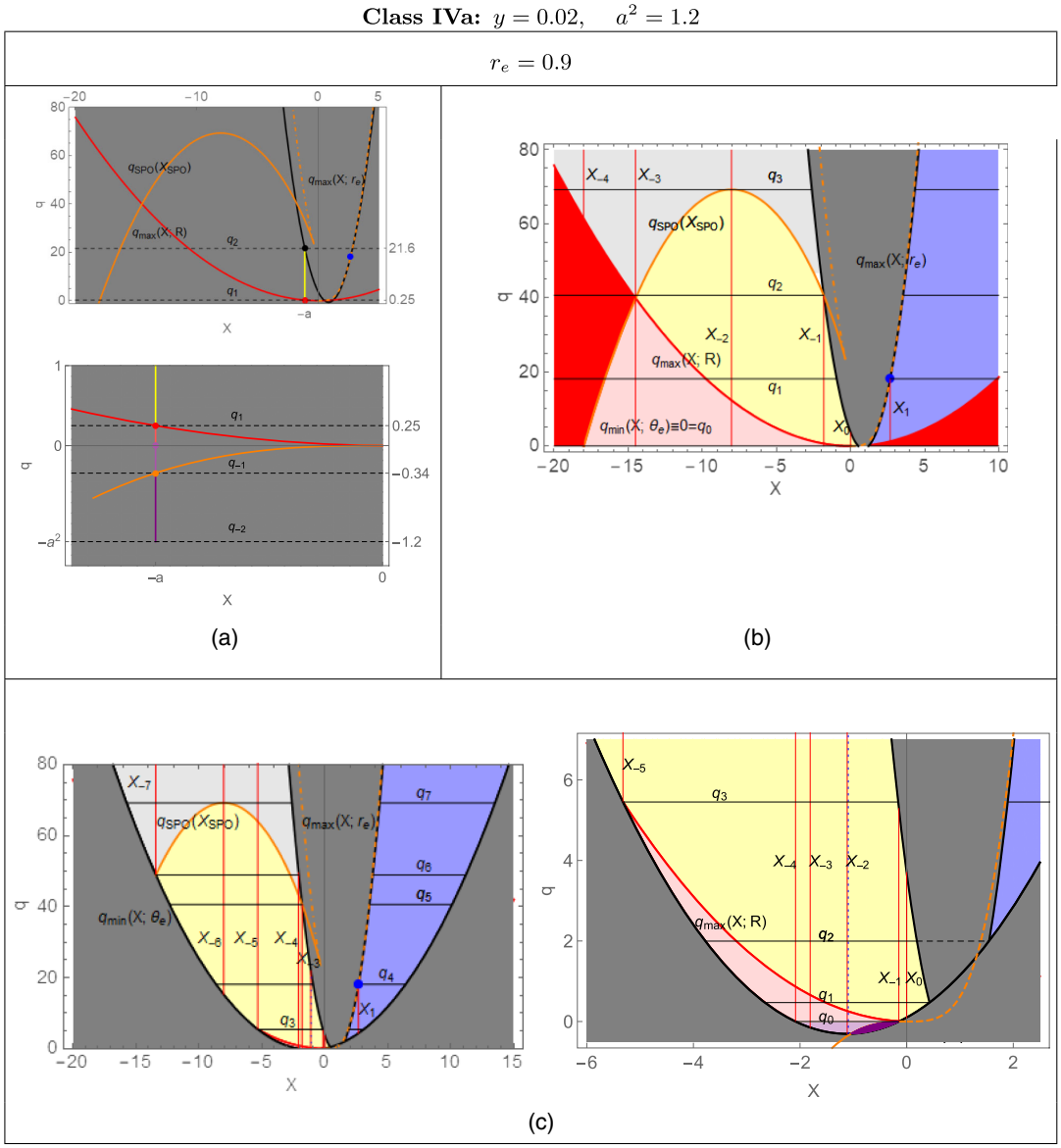


FIG. 11. Planes of motion constants (X-q) for the displayed latitudinal coordinates θ_e corresponding to special location on the spin axis, in the equatorial plane, and one general location in between. The chosen radial coordinate fulfils $r_e < r^{+}_{d(sph)}$, which corresponds to the region of the inner ergosphere with stable locally counterrotating SPOs with negative covariant energy $E < 0$. The coloring reflects different properties of the photon motion in the same manner as in Fig. 10: yellow parts distinguish the motion constants of photons escaping to $r \to +\infty$, either directly, or with one turning point if inward directed; light gray parts correspond to the captured photons with positive covariant energy E, blue parts correspond to the captured photons with negative covariant energy $E < 0$, dark gray parts mark the forbidden regions due to the reality conditions (37) and (38). For $q < 0$, the light purple part corresponds to photons entering the region of negative radii $r < 0$ and repelled back to positive radii, the dark purple part corresponds to photons escaping to $r \to -\infty$. The light red part corresponds to photons engrossed by the surface of the superspinar if they are initially directed inward or to escaping photons if they are initially directed outward. The rich red part corresponds to photons absorbed by the surface of the superspinar independent of their initial radial motion. Full orange curves designate the critical locus corresponding to the unstable SPOs, dashed orange curve corresponds to the stable SPOs with $E < 0$, dot-dashed curve corresponds to the stable SPOs with $E > 0$. The stable SPOs are always hidden for photons by the potential barrier, the only exception being the photon emitted with the motion constants highlighted by the blue point, which denotes the touch point of the displayed curves. It stays on the stable SPO with the current radius r_e. For $\theta_e = 0°$, the condition (37) implies $X = -a$, hence the allowed region has shrunk to an abscissa $X = -a$, $-a^2 \le q \le q_{\max}(-a; r_e)$. For a better understanding of the meaning of the motion constants, Figs. 12 and 20 explicitly show the photon directions corresponding to the prominence curves $X = \text{const}$ and $q = \text{const}$, marked here by the red vertical lines and the black horizontal lines. Note that with the same labels they can have different meanings for different latitudes θ_e. It should be emphasised that here, and in all other figures below, the surface of the superspinar at radius $\mathcal{R} = 0.1$ is considered. (a) $\theta_e = 0°$. (b) $\theta_e = 90°$. (c) $\theta_e = 60°$.

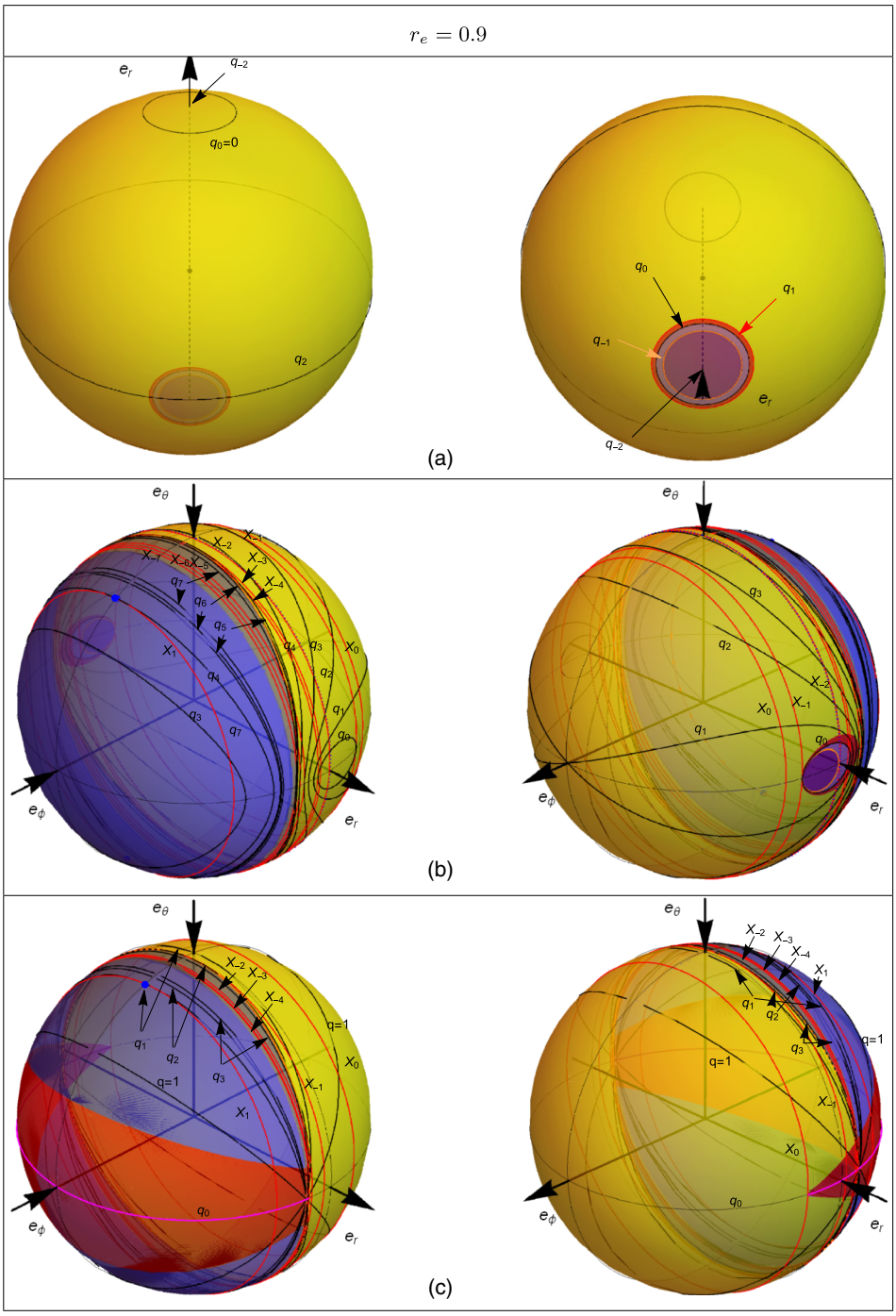


FIG. 12. Light escape cones represented by the spherical surface formed by the end points of the unit three-momentum vectors of the photons emitted by the source located at its center, corresponding to Fig. 11, i.e., the chosen radial coordinate r_e corresponds to the region of the inner ergosphere with negative covariant energy SPOs. The angle of view is apparent from the orientation of the local basis vectors. The meaning of the individual spherical sectors, curves, points, and their coloring can be read in the comments to Fig. 11. The light purple part corresponds to photons entering the region of negative radii $r < 0$ and repelled back to positive radii, the dark purple part corresponds to photons escaping to $r \to -\infty$. The boundary of this part represents photons that hit the ring singularity. For $\theta_e = 90°$ it shrinks to an arc in the equatorial plane depicted in magenta. A photon emitted in the direction represented by the blue point follows the stable SPO at the current coordinate $r_e = 0.9$. The actual observed shadow of the superspinar corresponds to the red region, while the shadow of the NS corresponds to the purple region; the dark purple represents a dark spot flanked by a circle (a) or sickle-shaped (b) bright fringe, represented by the light purple region. The shadow of the NS observed from the equatorial plane corresponds to the arc in magenta, but it is practically unobservable due to its infinitesimal thickness. The same conclusions apply to the shadows in the following images. (a) $\theta_e = 0°$. (b) $\theta_e = 60°$. (c) $\theta_e = 90°$.

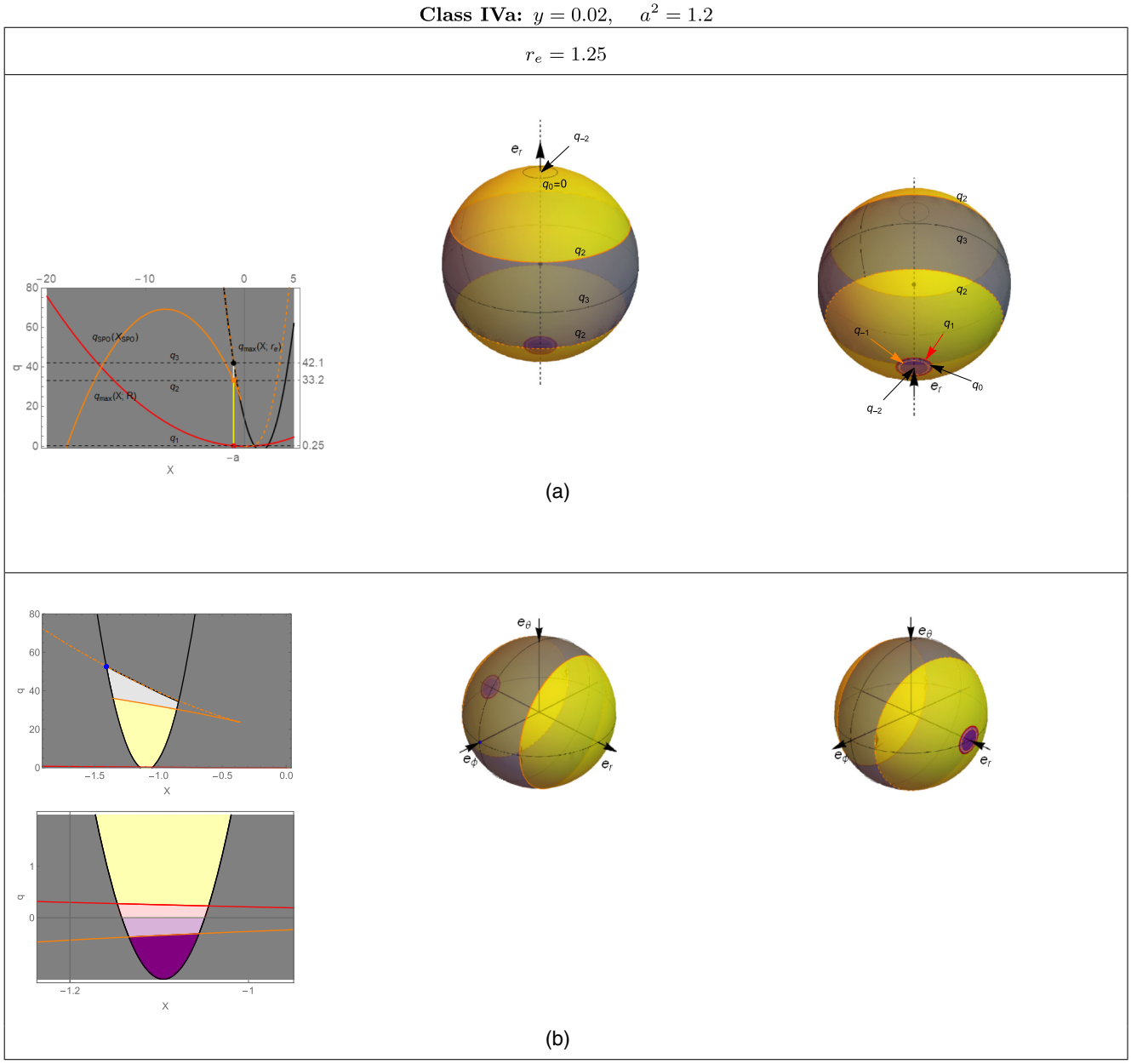


FIG. 13. Planes of motion constants (X-q) and affiliated LECs for $r^+_{d(SPO)} < r_e = 1.25 < r^+_{\rm pol}$ corresponding to the region of the inner ergosphere with stable locally counterrotating SPOs with positive covariant energy $E > 0$. In the left column, the full/dashed orange curves in the (X-q) plane for $q > 0$ correspond to the unstable/stable SPOs in the region $r > 0$, the full orange curves with $q < 0$ correspond to the unstable SPOs at $r < 0$. The meaning of the other curves follows from the captions. The yellow, light gray, light red, light purple, and purple region depicts the motion constants of the escaping, trapped, engrossed, repelled, and escaping to the other infinity photons, respectively. The coloring of the individual parts of the local celestial spheres of the emitter in the middle and right columns are chosen correspondingly. Dashed orange curves correspond to photons which reach some radial turning point and finally wind to the unstable polar SPO, while the full orange curve matches the photons that reach it directly. In the case $\theta_e = 0°$, the unstable SPO is the polar SPO at $r^-_{\rm pol} = 2.08$. The blue dot in the figures below represents a photon maintaining the current stable spherical orbit. (a) $\theta_e = 0°$. (b) $\theta_e = 2.40°$

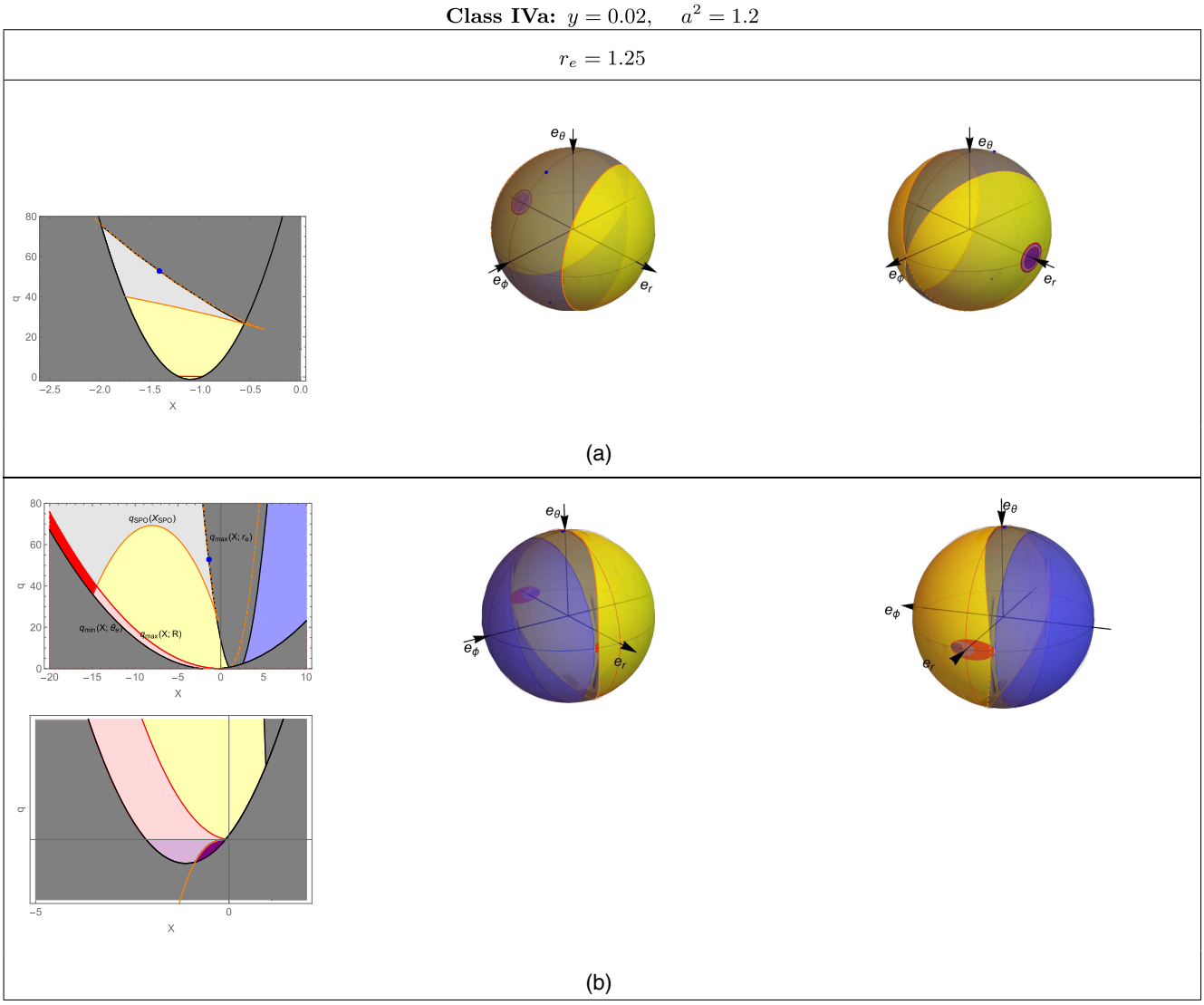


FIG. 14. Continuation of Fig. 13. The latitude in the upper figure is chosen so that a continuous transition of the shape of the trapped cone from the spherical belt to the spherical biangle can be seen, whose circular edges finally disconnect for $\theta_{\max(circ)}$. In the figure below, the source is inside the ergoregion $\theta_e > \theta_{\text{erg}} = 22.3°$, so there are photons detected with $E < 0$, corresponding to the blue regions on the (X-q) plane and on the celestial sphere. The light/rich red parts of the (X-q) plane correspond to engrossed photons reaching the superspinar surface directly/after passing through a radial turning point. The exact value of the latitude corresponds to the marginal case, when negative energy photons start to reach the superspinar surface. (a) $\theta_e = \theta_{\max(circ)} = 5.74°$. (b) $\theta_e = 66.28°$.

There is an outstanding region of the stable SPOs separated by the forbidden region in the spacetimes of class V (see Fig. 24), located under the radii $r_{\text{SPO}}^{\max}$, while the unstable orbits are in the range as in the classes VII and VIII. For emitters located at radii $r_e \lesssim r_{\text{SPO}}^{\max}$, the cone of trapped photons is delimited by a circle on the two-sphere, which occurs for latitude greater than a certain minimum, $\theta_e > \theta_{\min(circ)}$; for $\theta_e = \theta_{\min(circ)}$ this cone shrinks to an infinitesimal extent in the negative ϕ direction. For details about class V spacetime, see [28].

3. Position in the equatorial plane

In the most interesting case $r_e > r_{ph}^-$ the photons escaping toward the unstable SPOs always make an arc, which central angle ξ we investigate below. We also investigate the central angle η of the wedge-shaped shadow of the superspinar.

In the NS spacetimes, the ring singularity is reached by photons with motion constants $q = 0$ and $X_{\text{SPO}}(r_{ph}^-) \leq X \leq 0$. They are depicted in magenta as an arc in the equatorial plane.

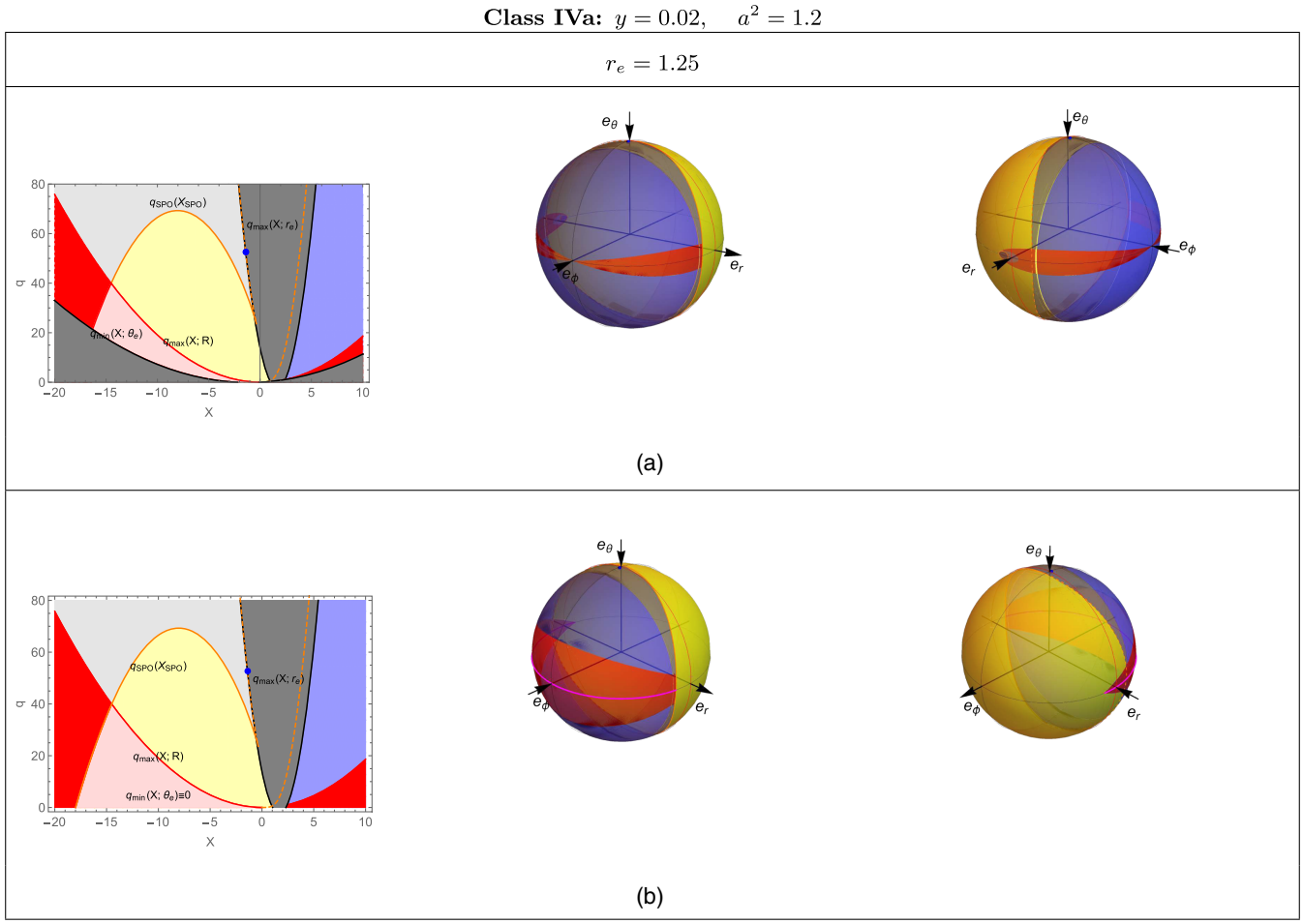


FIG. 15. Continuation of Fig. 14. The surface of the superspinar with increasing latitude toward the equatorial plane is successively impacted by the negative energy photons in both positive ($k^{(r)} > 0$) and negative ($k^{(r)} < 0$) locally measured radial directions. The meaning of the coloring is the same as in the previous figures. (a) $\theta_e = 72.86°$. (b) $\theta_e = 90°$.

F. Relation between the LECs and the superspinar shadows

The LECs constructed in the 3D mode presented below can also be used as a representation of the observer's local sky including the superspinar shadow. The construction of the shadows in the 3D mode would proceed in an analogous way using the analysis of photon motion above. Each incoming photon is associated with a point on the celestial sphere where the associated ray of light would pass through the sphere. Since with the same motion constants X, q, the incoming photon would pass through the sphere symmetrically with respect to the apparent image of the spin axis as related to the leaving photon, the emitter's LEC and the observer's celestial sky are symmetric overall. However, what appears to be symmetric from the center looks as if it were seen from the outside of the sphere. Hence, the below constructed 3D LECs can be viewed as the observer's star globes, depicting his celestial sky viewed from the point of his residence. This is the another advantage of this embodiment.

Whatever is shown on the LECs has a counterpart on the globe. For instance, the engrossed cones correspond to the superspinar shadow, since no photons with the appropriate motion constants can come to the observer from far distances from the corresponding directions. Furthermore, if there were no source of radiation in the directions of the trapped cones at $0 < r_e < r_{ph}^-$, the observer at $r_o = r_e$ should see a dark sky in directions that are axially symmetric with respect to the apparent image of the spin axis. On the other hand, if there were a source, e.g., hot matter from the accretion disk, there would be a large amount of radiation coming from these directions.

If the observer is located at $r = r_o > r_{ph}^-$, then he sees an arc as a part of the superspinar shadow. The arc itself is infinitesimally thin, but it is conspicuous due to the distortion of star patterns, which images lie along the arc, as the photons with motion constants $X \approx X_{\rm SPO}$, $q \approx q_{\rm SPO}$ emitted from the far stars must go many times around some unstable SPO corresponding to the arc before they reach the observer.

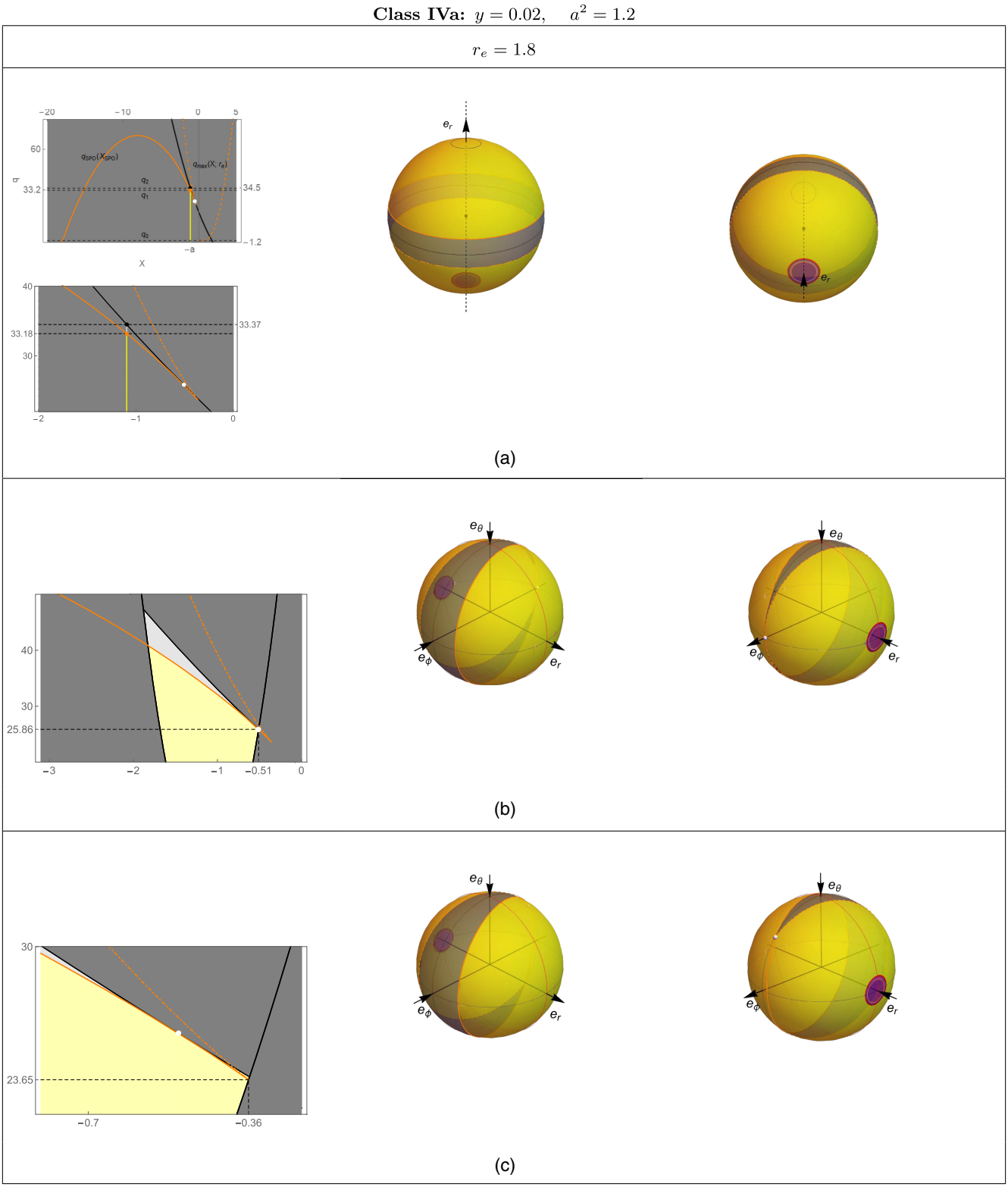


FIG. 16. LECs for $r_e = 1.8$ representing the case $r_{ms} < r_e < r^-_{\rm pol}$, corresponding to the region of the inner ergosphere with unstable corotating SPOs with positive covariant energy $E > 0$. For $\theta_e = 0°$ the there exist a trapped cone as in previous case. Since $r_e < r^-_{\rm pol}$, the trapped cone exists for any latitude. A photon emitted in the direction represented by the white point follows the unstable SPO at the current coordinate r_e. As a result of emitter's radial position, the lower SPOs are reached only by inward photons, so the full orange curve exceeds the dashed one and for latitude $\theta_{\max(circ)}$ it disconnects and an arc appears. (a) $\theta_e = 0°$. (b) $\theta_e = 6.36°$. (c) $\theta_e = \theta_{\max(circ)} = 8.29°$.

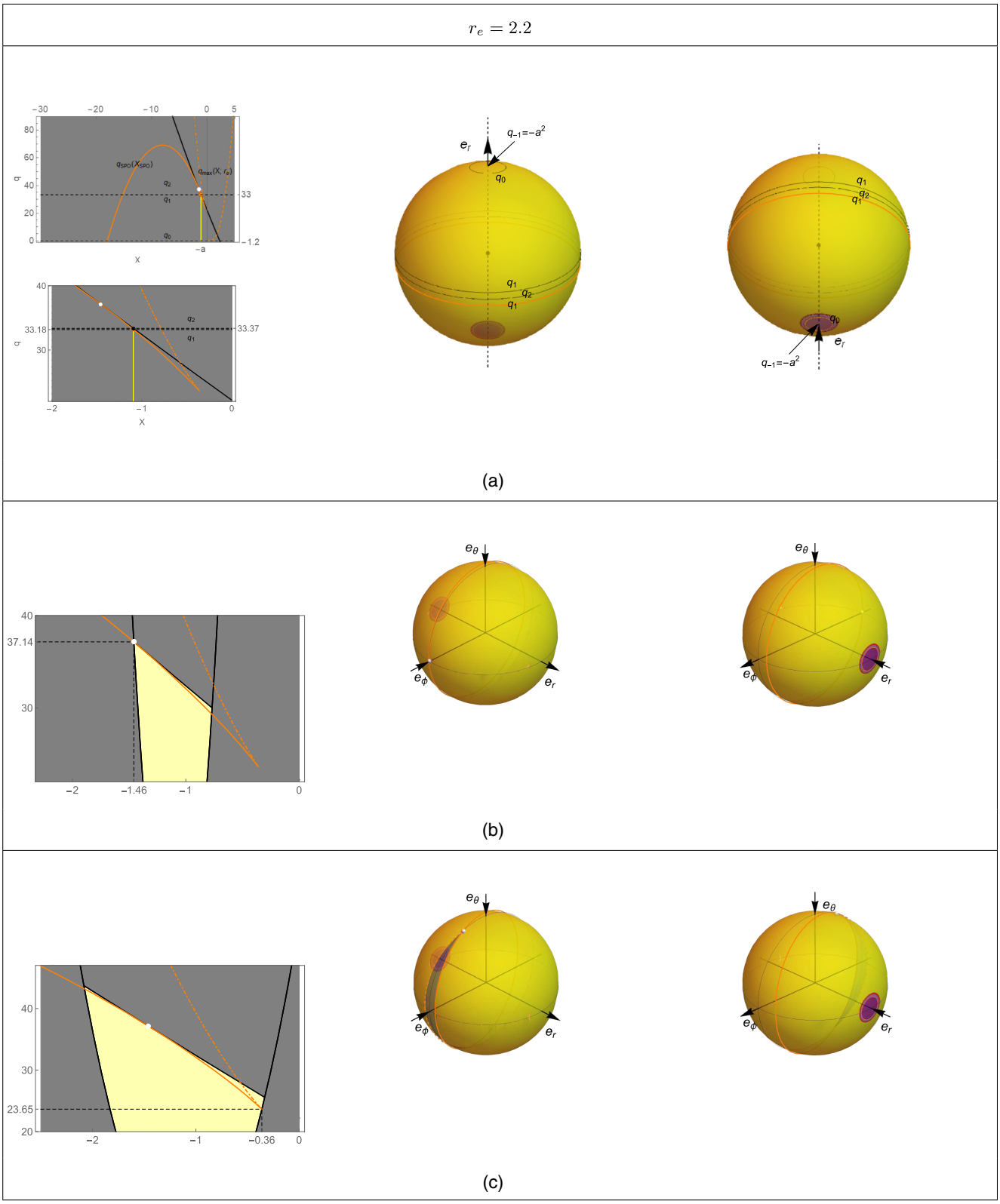


FIG. 17. LECs for $r_e = 2.2$ representing the case $r^-_{\rm pol} < r_e < r^+_{\rm erg}$, corresponding to the region of the inner ergosphere with unstable counterrotating SPOs. There are no trapped cones on the spin axis, since all photons emitted to the polar SPO are directed inward. For $\theta_e = 3.32°$ there is a photon directed on the actual unstable SPO, represented by the white dot. For $\theta_e > \theta_{\max(circ)} = 8.29°$ the orange curve disconnects and an arc is formed. (a) $\theta_e = 0°$. (b) $\theta_e = 3.32°$. (c) $\theta_e = \theta_{\max(circ)} = 8.29°$.

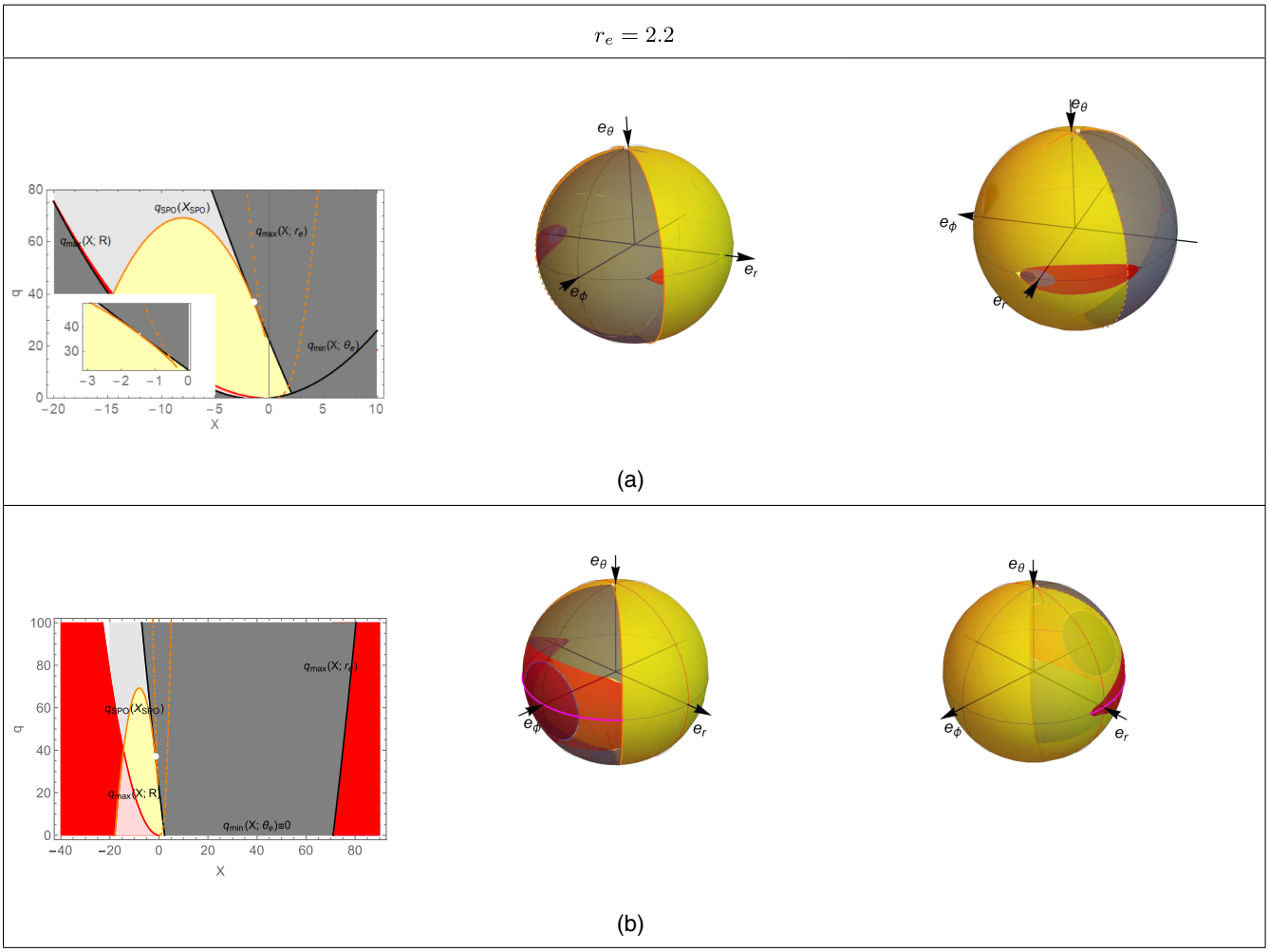


FIG. 18. Continuation of Fig. 17. Since $r_e < r_{ph}^-$ there is a trapped cone for sufficiently large latitudes. Furthermore, due to the position of the emitter in the ergosphere, there is a marginal latitude $\theta_{\rm erg}(r) = 69.5°$, so that for $\theta_e > \theta_{\rm erg}(r)$ there is a flux of trapped photons with $E < 0$. In the figure below, all photons with negative energy are absorbed by the surface of the superspinar, which is shown by the blue region being covered by the red. (a) $\theta_e = 65°$. (b) $\theta_e = 90°$.

When constructing the shadow of a superspinar, we consider for comparison the situation where there is a naked singularity in its position. Apart from the arc, there is a direct silhouette of the NS given by the null geodesics connecting the ring singularity and its interior with the observer. The dark spot is seen in the directions where the photons escape to the other infinity (at negative radii sheet), its boundary is determined by photons with the motion constants $X_{\rm SPO}(r_{\rm SPO} < 0), q_{\rm SPO}(r_{\rm SPO} < 0)$. The null geodesics from the remaining part have a turning point in region $r < 0$, so if there is a source of illumination, the region on the observer's sky corresponding to photons coming from the directions between the ring singularity and the boundary of the dark spot could be visible.

G. Relevance of the polar SPOs with construction of the LECs/shadows

From the above we can summarize that the relevance of the polar SPOs with the construction of the LECs and superspinar/NS shadows is as follows. If a spacetime is endowed with the polar SPOs, then the critical locus $q_{\rm SPO}(X_{\rm SPO})$ in the corresponding $(X\text{-}q)$ plane of motion constants is defined for $X_{\rm SPO} = -a$ $(\ell_{\rm SPO} = 0)$. It then intersects the degenerated graph of the function $q_{\rm min}(X; \theta_e \to 0)$ defined by the relations (45) and (46). As the photon emitter moves away from the spacetime symmetry axis, i.e., as θ_e grows, the parabola $q_{\rm min}(X; \theta_e)$ cuts an arc at the critical locus. According to the radial position of the emitter relative to the outer unstable

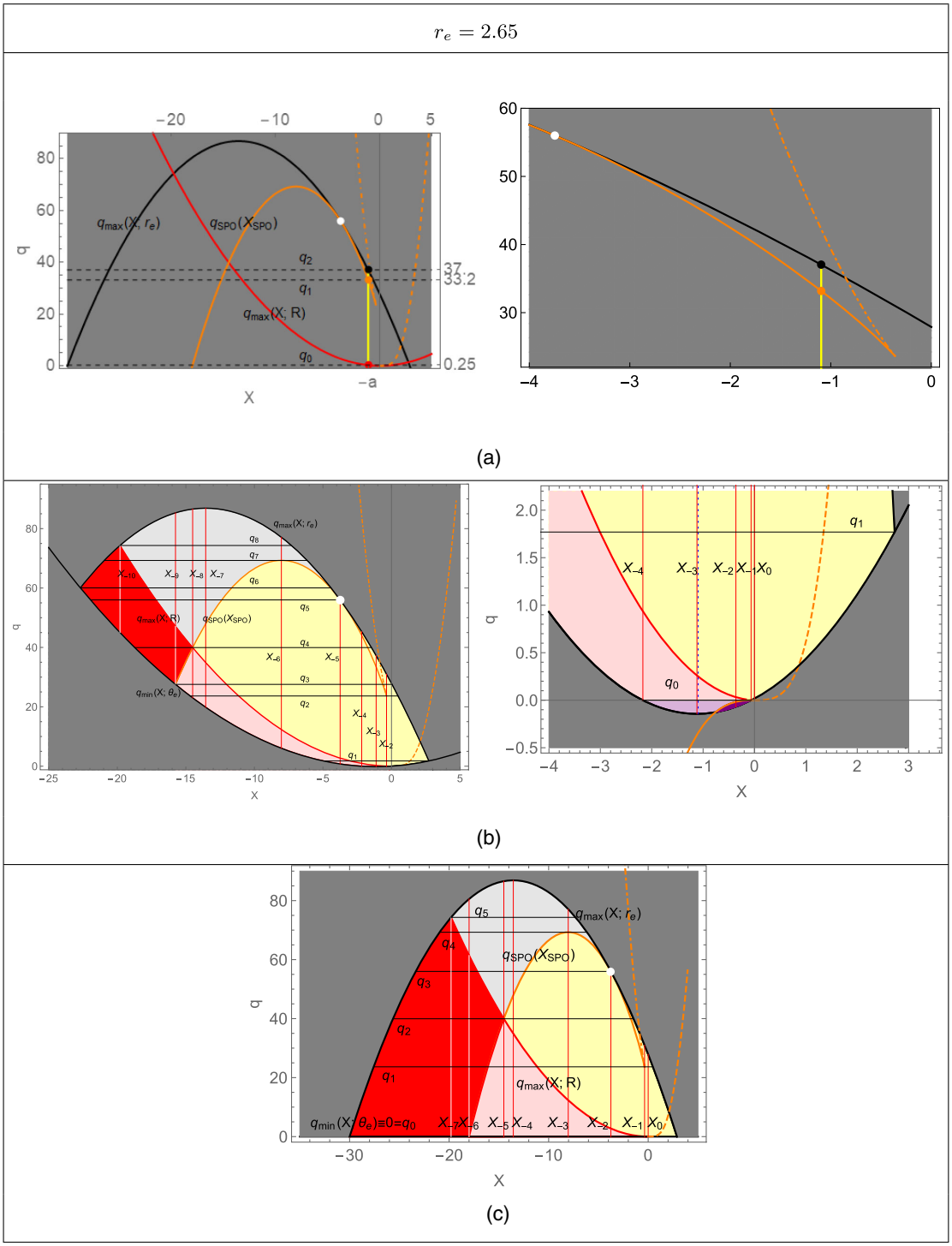


FIG. 19. Planes of the motion constants (X-q) for the case $a^2 = 1.2$, $y = 0.02$, $r_e = 2.65$. The latitudinal coordinates correspond to special cases on the spin axis, in the equatorial plane, and in the general position. The radial coordinate corresponds to case $r^+_{\rm erg} < r_e < r^-_{ph}$, i.e., the region of unstable counterrotating SPOs with positive covariant energy $E > 0$ outside the ergosphere, hence, the function $q_{\rm max}(X; r_e)$ is now concave. The label X_{-3} in the detail on the right denotes two close lines, namely, $X = -1.116$ (red), which is local minimum of $q_{\rm min}(X;\theta)$, and $X = a = -1.095$, i.e., $\ell = 0$ (blue, dotted). (a) $\theta_e = 0°$. (b) $\theta_e = 70°$. (c) $\theta_e = 90°$.

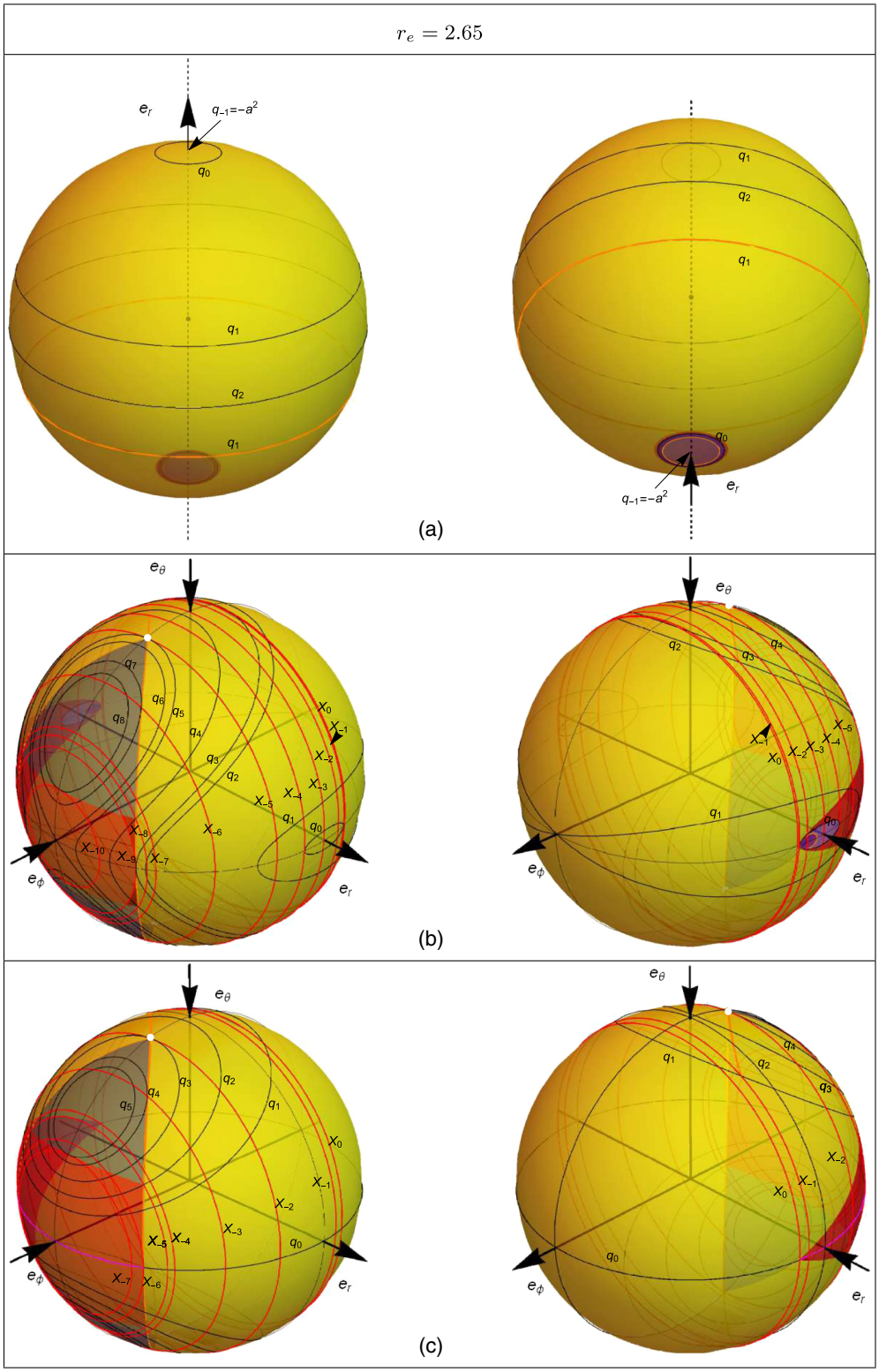


FIG. 20. Light escape cones associated with the planes of motion constants (X-q) in Fig. 19. (a) $\theta_e = 0°$, (b) $\theta_e = 70°$, and (c) $\theta_e = 90°$.

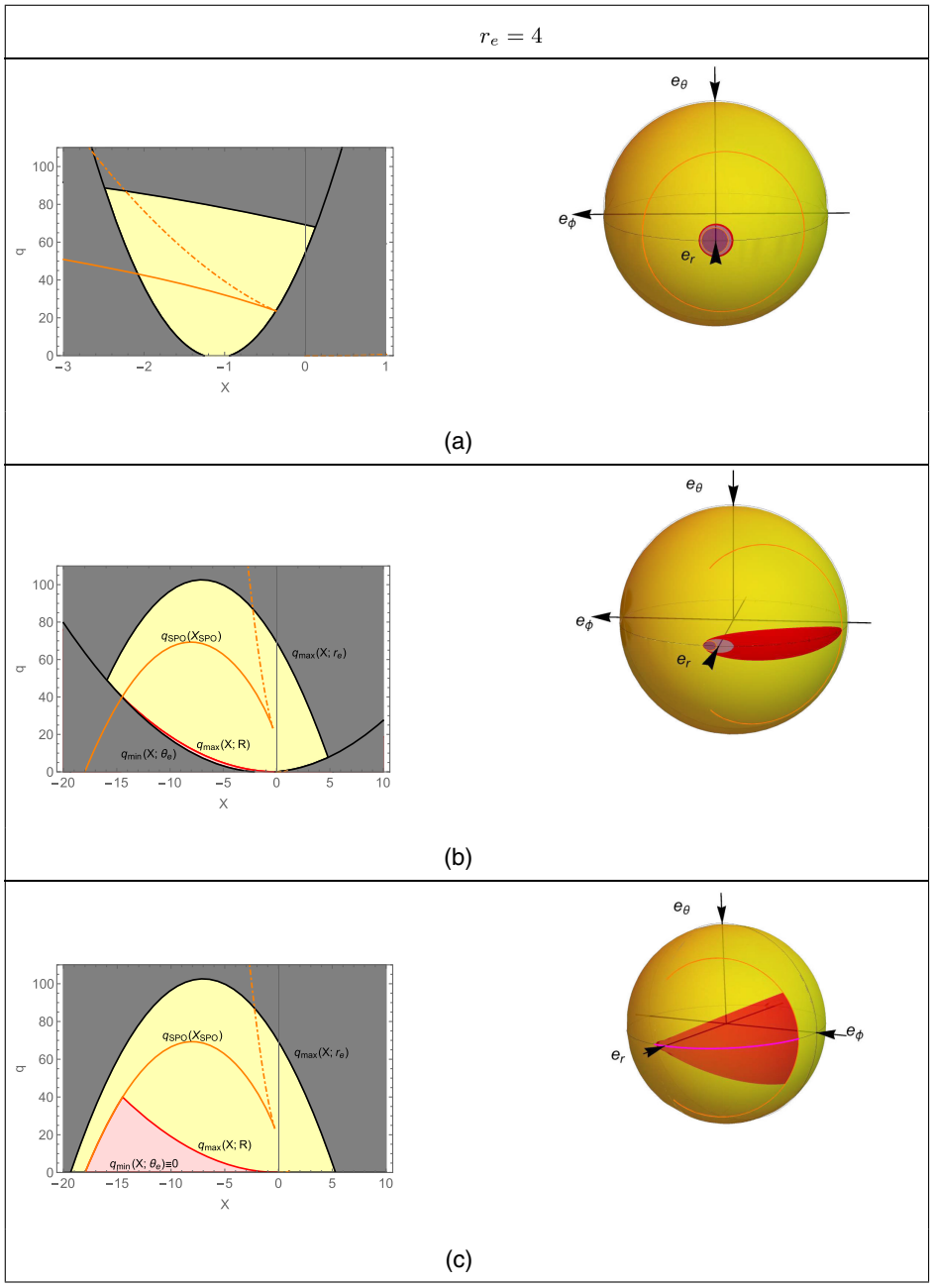


FIG. 21. LECs for the case $r_{ph}^- < r_e < r_{\rm erg}^-$, i.e., the region outside the ergosphere and SPOs and close to the static radius at $r_s = 3.68$. There are no trapped cones and an arc appears for latitudes large enough, $\theta_e > \theta_{\max(circ)} = 8.29°$. The middle figure introduces the limiting angle $\theta_{\max(ell)}$, for which the cone of absorbed photons is still elliptical. In the equatorial plane in the figure below, this cone transforms into a wedge shape. The cone of light escaping to $r \to -\infty$ and the repelled cone degenerate into the shape of a line in the equatorial plane, shown in magenta. (a) $\theta_e = \theta_{\max(circ)} = 8.29°$. (b) $\theta_e = \theta_{\max(ell)} = 64.39°$. (c) $\theta_e = 90°$.

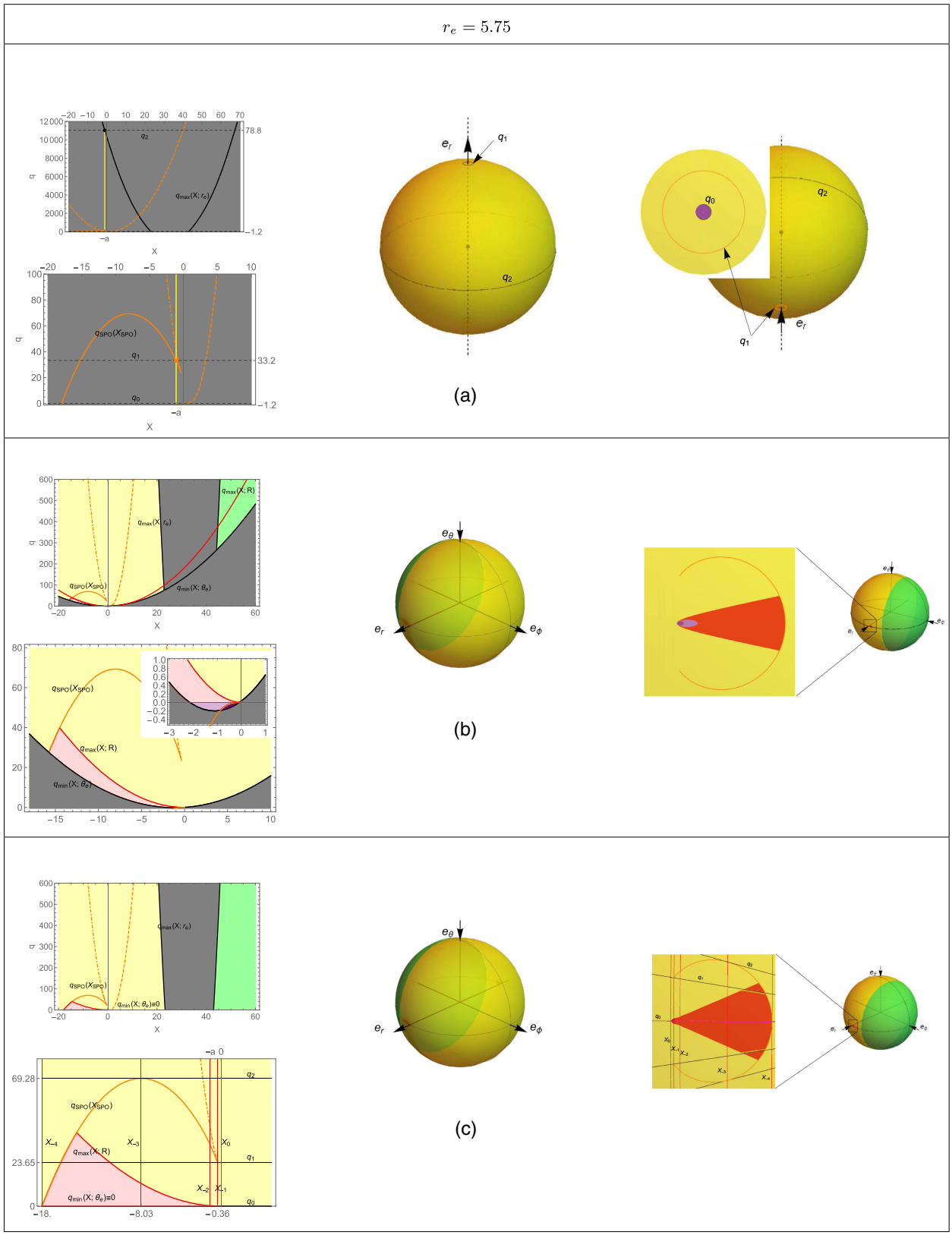


FIG. 22. LECs for the case $r_{\text{erg}}^{-} < r_e < r_c$, corresponding to the cosmological ergosphere, where the photons with negative locally measured azimuthal component $k^{(\phi)} < 0$ of their four-momentum and with negative covariant energy $E < 0$ occur. These photons correspond to the regions depicted in green. The meaning of the remaining coloring is the same as in the previous case. (a) $\theta_e = 0°$. (b) $\theta_e = 70°$. (c) $\theta_e = 90°$.

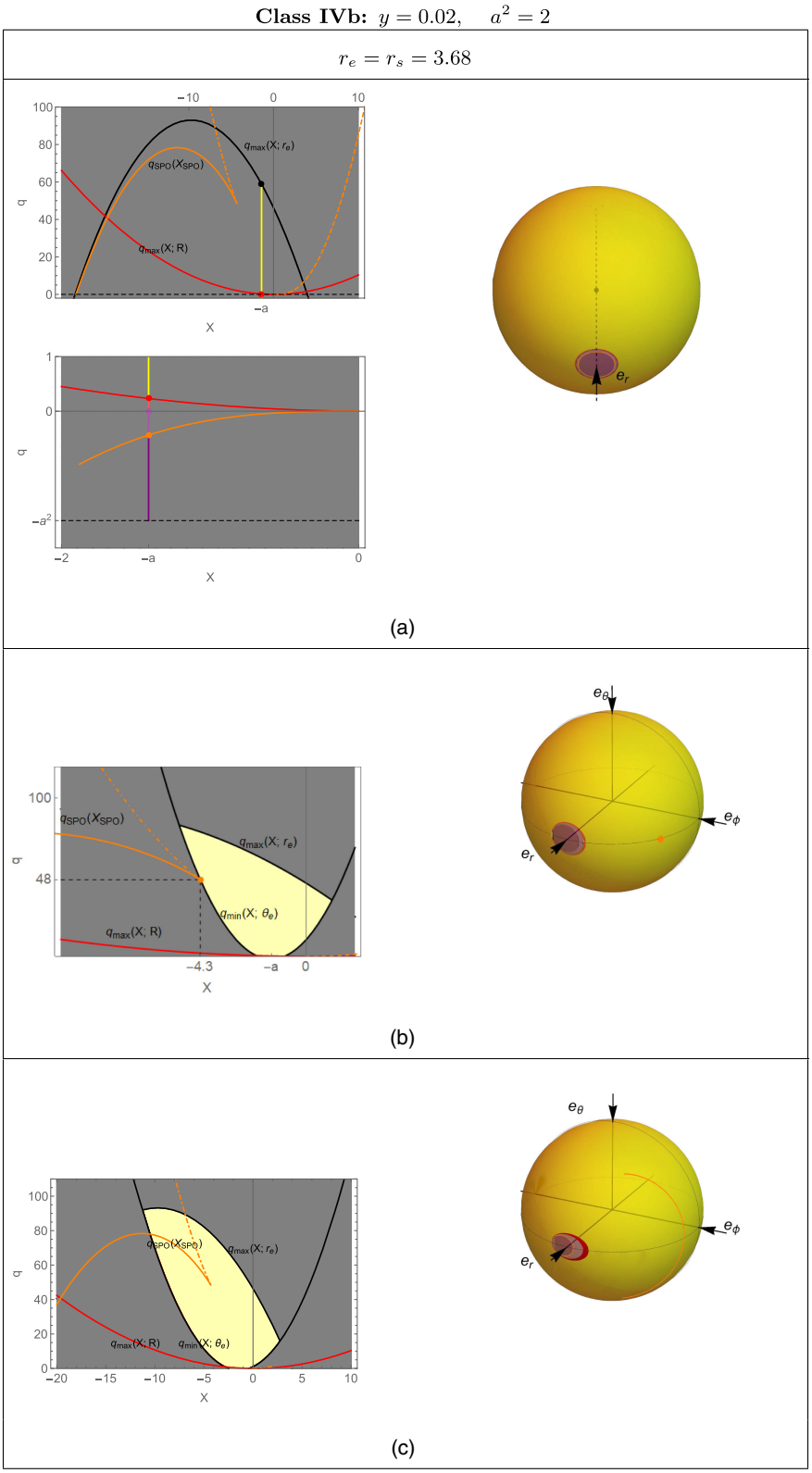


FIG. 23. LECs in spacetime with DRB and no polar SPOs. The radii of the observers are chosen to be at the static radius $r_s = 3.68$. Since there are no polar SPOs, there is neither a trapped cone nor critical locus for an observer located on the polar axis, but it appears as an arc for $\theta_e > \theta_{\min(arc)} = 21.79°$. Otherwise, the appearance of the LECs for $\theta_e > \theta_{\min(arc)}$ remains similar to the case of the spacetimes of the class IVa and angles $\theta_e > \theta_{\max(circ)}$ as presented in the figures above. (a) $\theta_e = 0°$. (b) $\theta_e = \theta_{\min(arc)} = 21.79°$. (c) $\theta_e = 45°$.

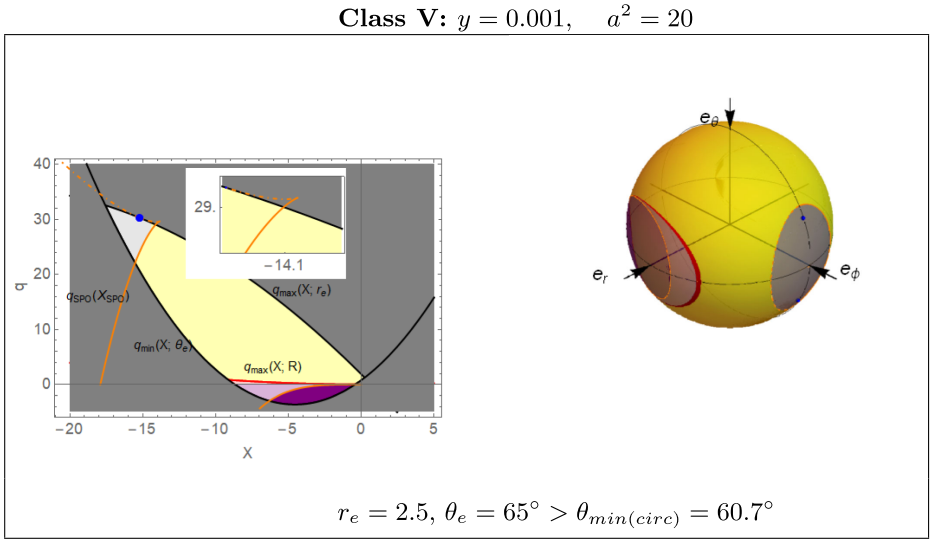


FIG. 24. LEC in the case of class V spacetime with small region of unusual bound orbits with $X_+ < X < 0$ and $E > 0$ [see the detail in Fig. 10(b)]. It causes that for $r_e \lesssim r^{\max}_{\rm SPO}$ the critical locus $X_{\rm SPO}(q_{\rm SPO})$ is partly hidden in the forbidden region (detail in the figure left). As a consequence, there exists a minimal angle $\theta_{\min(circ)}$, such that for $\theta_e > \theta_{\min(circ)}$, here $\theta_{\min(circ)} = 60.7°$, a light cone of the trapped photons appears, which is bounded by a circlelike-shaped rim on the sphere (figure right). For $\theta_e = \theta_{\min(circ)}$ this cone shrinks to an infinitesimal extent directed in the negative ϕ direction.

spherical photon orbit $r^-_{\rm pol}$ and the equatorial circular photon orbit r^-_{ph}, this may cause a spherical belt or biangle to form on the emitter's celestial sphere, where the emitted photons are trapped, or just a circle separating the two cones of the escaping photons (see, e.g., Figs. 14, 16, and 17 and the comments). Then there exists a critical angle $\theta_{\max(circ)}$ for which the circle disconnects and an arc appears [cf., e.g., Figs. 18(a) and 18(b) or Figs. 21(b) and 21(c)].

On the other hand, if there are no polar SPOs in the spacetime, then the intersection of the graph of the function $q_{\min}(X;\theta_e)$ with the critical locus appears after a certain minimum angle is attained. We denoted it $\theta_{\min(arc)}$ in the spacetimes of class IVb or VIb, since for $\theta_e > \theta_{\min(arc)}$ an arc occurs (see Fig. 23), and $\theta_{\min(circ)}$ for the spacetimes of class V, since for $\theta_e > \theta_{\min(circ)}$ a circlelike-shaped boundary of the trapped cone occurs (see Fig. 24). Then there is neither a belt nor an arc for $\theta_e < \theta_{\min(arc/circ)}$.

V. ASTROMETRIC OBSERVABLES

We can now relate our constructions to the notion of the KdS superspinar shadow to the astrophysical context, namely, to coordinates used by nearly static observers at a large distance from the superspinar.

A. Celestial coordinates

In the following we shall focus on the detail of the observer's celestial sphere containing the superspinar shadows for an observer at a large distance. For our purposes it is convenient to introduce the celestial coordinates $\tilde\alpha, \tilde\beta$, which are related to the directional angles α, β by the relations

$$\tilde\alpha = -\sin\alpha\sin\beta = -k^{(\phi)}/k^{(t)},$$
$$\tilde\beta = \sin\alpha\cos\beta = k^{(\theta)}/k^{(t)}. \tag{80}$$

These angles are, as in the Kerr case, $y = 0$, proportional to $1/r$, however, here we cannot use the usual formula (see, e.g., [37,43,85])

$$\tilde\alpha = \lim_{r_o\to\infty}(-r_o k^{(\phi)}/k^{(t)}), \tag{81}$$

$$\tilde\beta = \lim_{r_o\to\infty}(r_o k^{(\theta)}/k^{(t)}), \tag{82}$$

as we cannot place the observer at arbitrary distances due to the presence of the cosmological horizon at $r = r_c$. We shall deal with a distant observer, by which we mean an observer located outside the region of the SPOs, i.e., at $r^-_{ph} < r_o < r_c$. Following our previous work [77], we choose the family of LNRFs near the static radius in Kerr–de Sitter spacetimes, since they exhibit characteristics similar to those of static observers at large distances in asymptotically flat spacetimes [81,82].

B. Observable quantities

In Fig. 25 we present the observables that we shall investigate below. These observables depend on the spacetime parameters and both the latitudinal and radial position of the observer. In Figs. 26–28 we show the radial dependence of some observables for several values of the dimensionless cosmological constant and for two

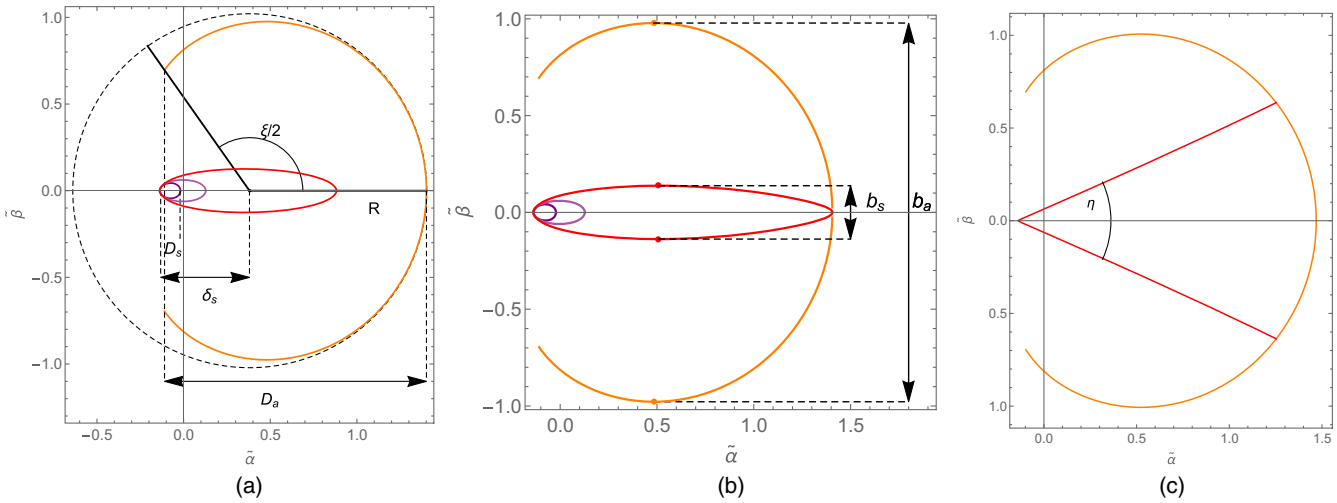


FIG. 25. The observables investigated below: (a) the radius R of the osculating circle, approximating the light arc, lining either the shadow of the superspinar with surface at radius $\mathcal{R} = 0.1$ (red curve), or the NS shadow, consisting of the dark spot corresponding to photons escaping to the region of negative radii (darker purple curve) and photons entering the $r < 0$ region but repelled back to the $r > 0$ region (region between the darker and the light purple curve, which corresponds to photons terminating at the ring singularity); the diameter D_s of the dark spot, the diameter D_a of the arc, the distance δ_s between the left end point of the spot and the center of the osculating circle, and the central angle ζ of the arc; (b) the "vertical" diameter b_s of the largest elliptical shadow of the superspinar just reaching the arc with vertical diameter b_a, observed at some critical latitude $\theta_{\mathrm{max}(ell)}$ to be investigated below; (c) the angle η of the wedge representing the shadow of the superspinar observed from the equatorial plane. Note that all the quantities are measured in radians.

representative spin parameters corresponding to the class IVa and VIa KdS superspinar (NS) spacetimes, i.e., both with DRB but the former with the polar SPOs, the latter without the polar SPOs. We shall not deal with the spacetimes with RRB, as they are not astrophysically relevant due to implausibly high values of the dimensionless cosmological constant.

The fixed value $a^2 = 1.2 \approx a^2_{\mathrm{crit}}$ was chosen as a "medium" spin value of class IVa KdS spacetimes, allowing the existence of the polar SPOs. In such spacetimes, the shadow consists of the dark spot surrounded by a full "circle," observed for latitudes θ_o smaller than some critical maximum $\theta_{\mathrm{max}(circ)}$ allowing the appearance of the circle. For $\theta_o > \theta_{\mathrm{max}(circ)}$, the circle splits into an arc (see Fig. 21). The dependence of the angle $\theta_{\mathrm{max}(circ)}$ on the spacetime parameters a^2, y is shown in Fig. 30(a). The central angle ξ of the arc decreases with latitude θ_o and it reaches its minimum value ξ_{min} for some latitude $\theta_{\xi(\mathrm{min})}$, $0 < \theta_{\xi(\mathrm{min})} < 90°$, from which it grows slightly to its local maximum ξ_{eq} in the equatorial plane $\theta_o = 90°$ [see Fig. 29(a)]. The dependence of the angles $\theta_{\xi(\mathrm{min})}$, ξ_{min} on the spacetime parameters a^2, y is shown in Figs. 30(b)–30(d).

The fixed value $a^2 = 2$ corresponds to a spacetime without the polar SPOs. In such spacetimes there is no arc for latitudes less than some critical minimum $\theta_{\mathrm{min}(arc)}$ close to the spin axis, $\theta_o < \theta_{\mathrm{min}(arc)}$. For $\theta_o = \theta_{\mathrm{min}(arc)}$ a degenerate arc appears which appears as a spot [see Fig. 23(b)], i.e., with the central angle $\xi = 0$. For $\theta_{\mathrm{min}(arc)} < \theta_o$ the arc increases and its central angle grows monotonically to its maximum value ξ_{eq} in the equatorial plane $\theta_o = 90°$ [see Fig. 29(b)]. The dependence of the angles $\theta_{\mathrm{min}(arc)}$, ξ_{eq} on the spacetime parameters a^2, y is shown in Fig. 31.

In Fig. 32 we plot the dependence of the angle $\theta_{\mathrm{max}(ell)}$ on the spacetime parameters, which represents the maximum angle at which the observer can still see the elliptical shadow of the superspinar, which appears to be the largest [see Fig. 21(b)].

Figure 33 shows the dependence of the ratio of the quantities b_s, b_a, defined in Fig. 25(b), on the spacetime parameters for different radii of the superspinar surface $\mathcal{R}$.

The maximum peak angle η of the wedge-shaped shadow of the superspinar as seen by the observers in the equatorial plane is plotted as a function of the spacetime parameters in Fig. 34.

Figures 35 and 36 show the comparison of the superspinar shadows versus the NS shadows for a fixed value of the parameter a^2/y while varying the parameter y/a^2, respectively. We used the radial position on the static radius $r_o = r_s$, as its vicinity represents the best local approximation to the asymptotically flat region of the Kerr spacetime [82,83], and two representative observer latitudes, i.e., 60°, representing the position outside the equatorial plane, and the second latitude corresponding to the equatorial plane.

In Figs. 37 and 38 we investigate the influence of the observer latitude on the observed images of the superspinar (NS) for both fixed spacetime parameters a^2, y as seen on

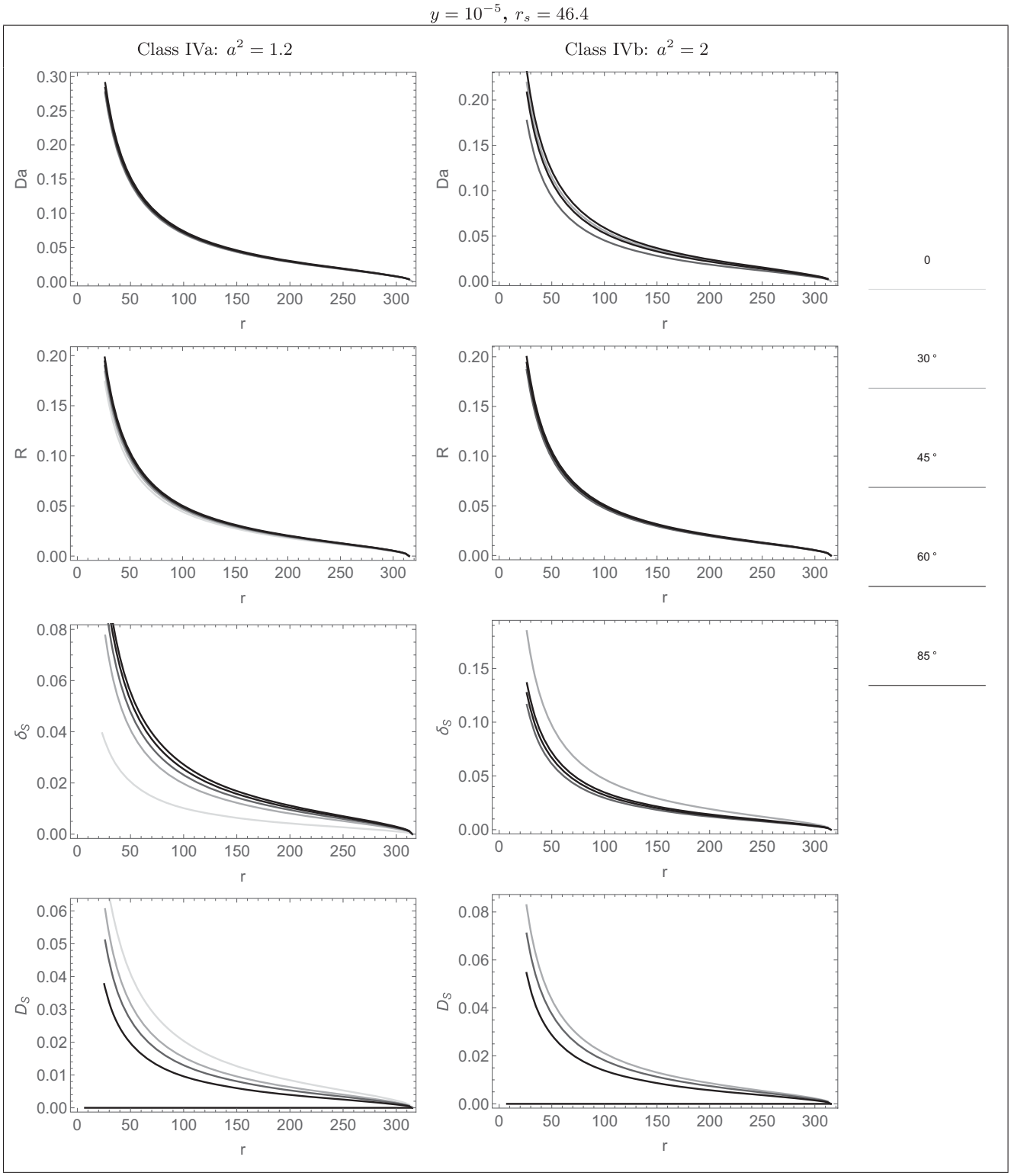


FIG. 26. Comparison of the dependence of the observables on the radial position of an observer for some selected values of its latitudinal position for the case of the KdS spacetimes with (left) and without (right) polar SPOs.

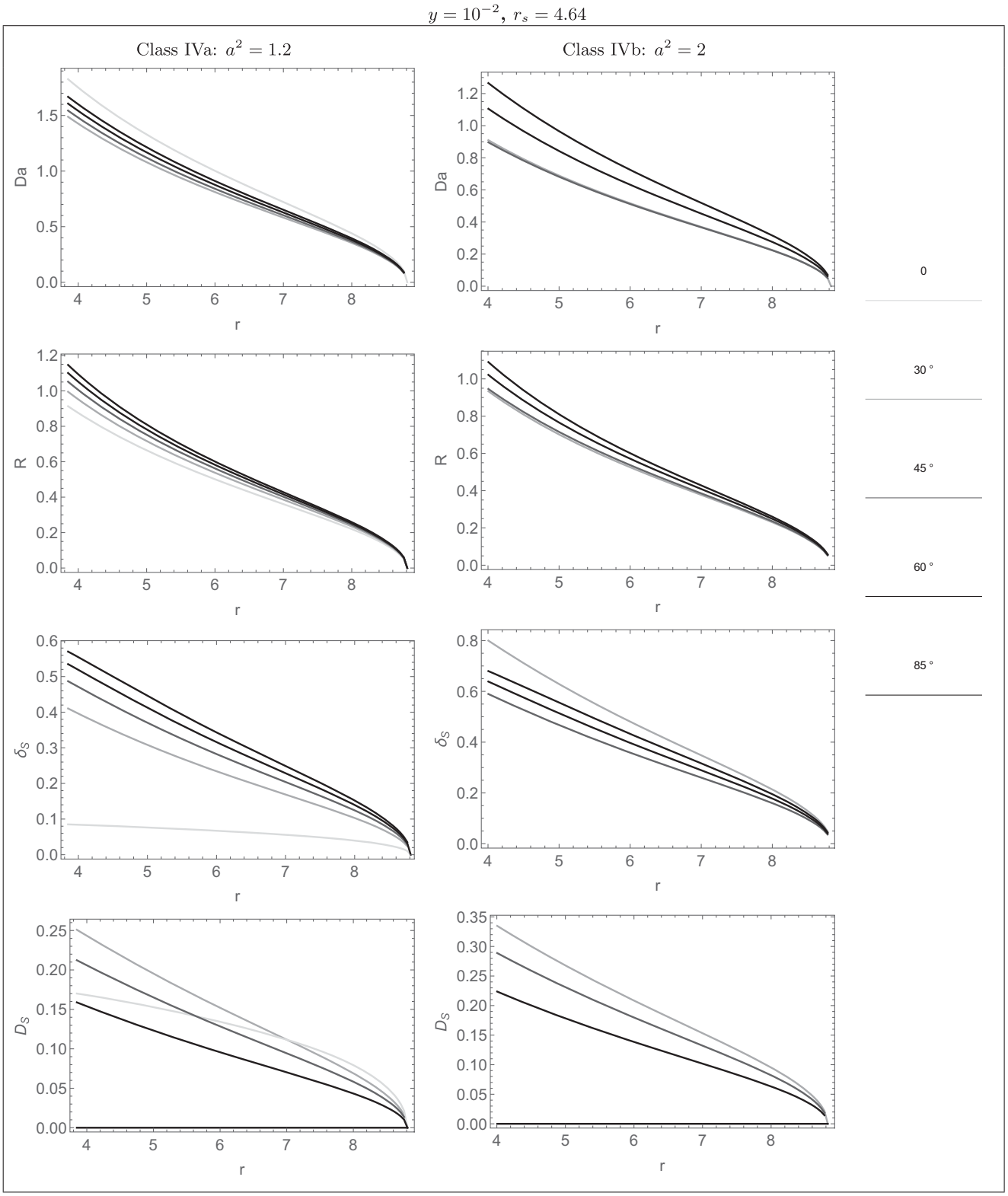


FIG. 27. Comparison of the dependence of the observables on the radial position of an observer for some selected values of its latitudinal position for the case of the KdS spacetimes with (left) and without (right) polar SPOs.

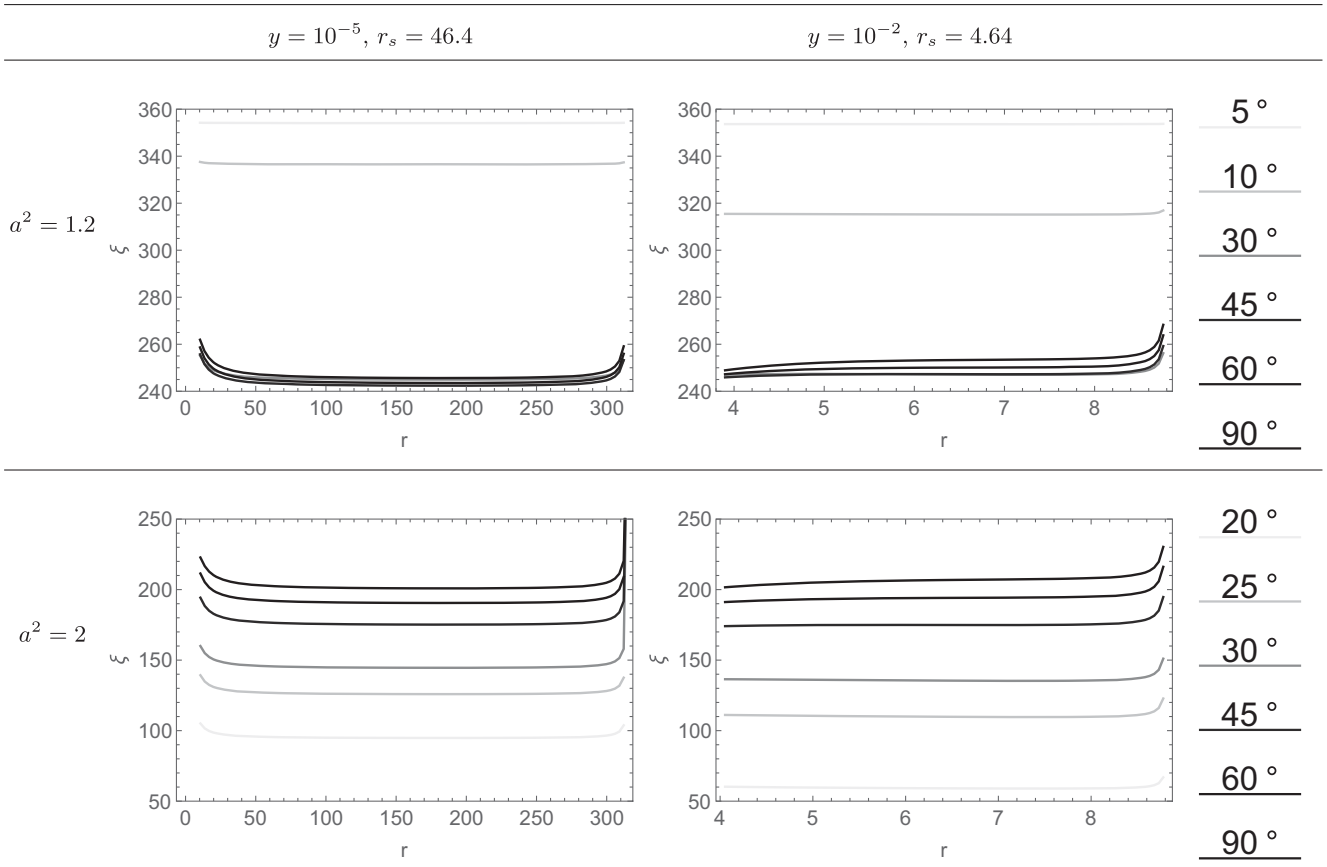


FIG. 28. Dependence of the central angle ξ of the arc on the radial position of the observer for some selected values of his latitudinal position. The top row corresponds to the NS spacetimes with polar SPOs, hence for $\theta_o < \theta_{\max(circ)}(y, a^2)$ it is $\xi(r) = 360°$. Here $\theta_{\max(circ)}(10^{-5}, 1.2) = 9.74°$, $\theta_{\max(circ)}(10^{-2}, 1.2) = 9.04°$. The bottom row corresponds to the NS spacetimes without polar SPOs, hence for $\theta_o < \theta_{\min(arc)}(y, a^2)$ there is no arc. Here $\theta_{\min(arc)}(10^{-5}, 2) = 15.5°$, $\theta_{\min(arc)}(10^{-2}, 2) = 18.5°$.

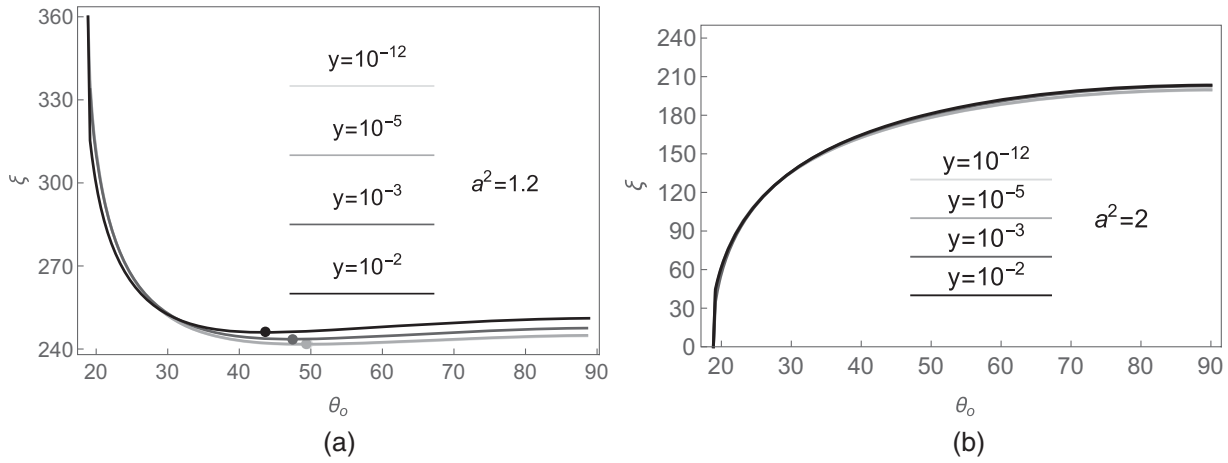


FIG. 29. Comparison of the dependence of the central angle ξ on the latitude of the observer at the appropriate static radius $r_s(y)$ for some selected spacetimes with polar SPOs (a) and for spacetimes with the same cosmological parameter y but different spin parameter a^2, corresponding to spacetimes with no polar SPOs (b). It can be seen that in the case of spacetimes with polar SPOs there is a certain local minimum $\xi_{\min}$ for a certain angle $\theta_{\xi(\min)}$ (highlighted by dots), while in the case of spacetimes without polar SPOs the angle ξ increases monotonically up to the value ξ_{eq} corresponding to the latitudinal coordinate in the equatorial plane $\theta = 90°$. The magnitude of both angles θ and ξ is plotted in degrees on both axes.

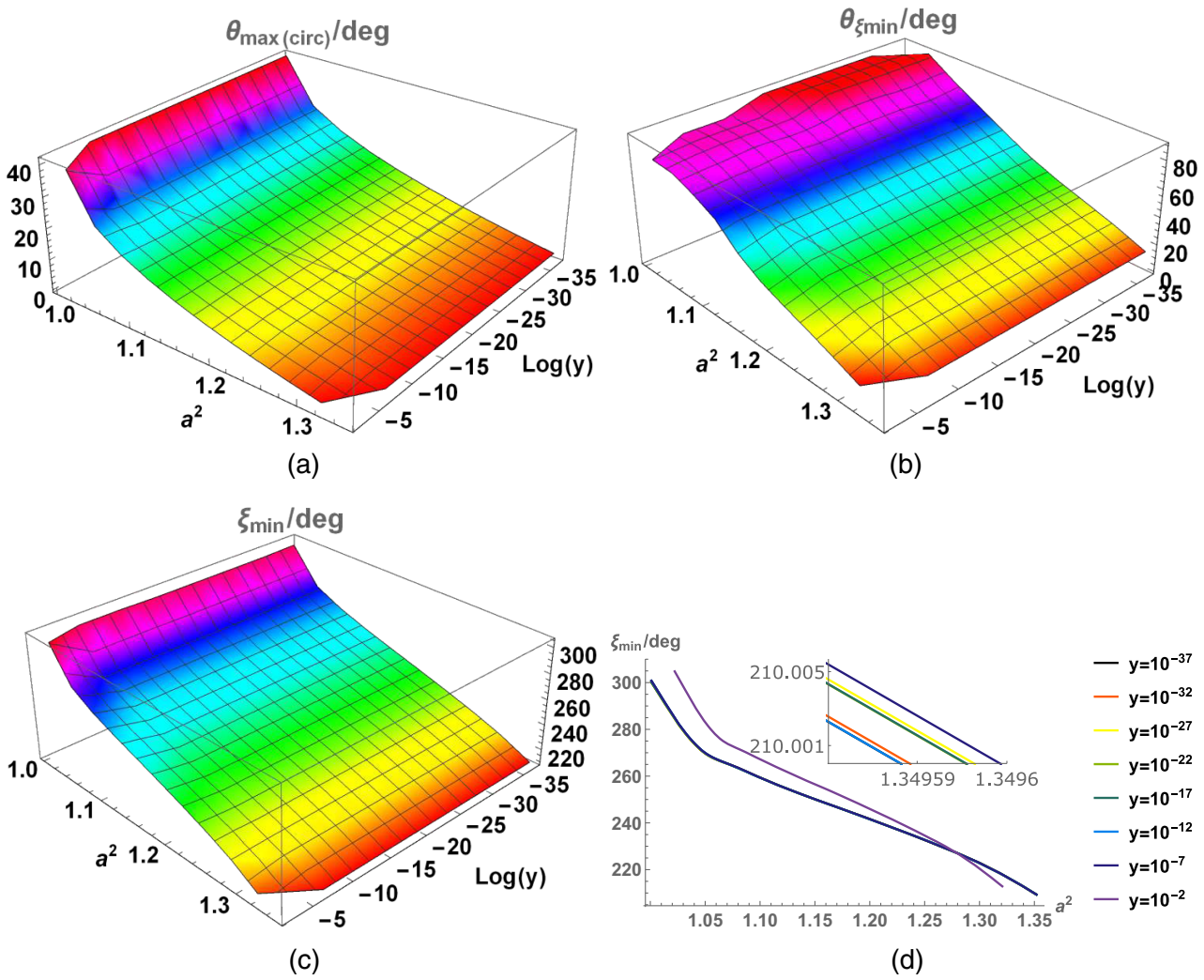


FIG. 30. Dependence of the characteristic angles on the spacetime parameters corresponding to class IVa: (a) the maximum angle $\theta_{\max(circ)}$ under which the observer can see the light circle lining the superspinar/NS shadow, as described, e.g., in Fig. 21(b); (b) the observer's latitude $\theta_{\xi(\min)}$ at which he sees the local minimum $\xi_{\min}$ of the angle ξ of the arc [cf. Fig. 29(a)]; (c) the value $\xi_{\min}$ of the minimum angle ξ of the arc; (d) slices of the 3D graph in (c) through the planes of the constant parameter y for some appropriate values. For $y \ll 10^{-2}$ the curves almost overlap.

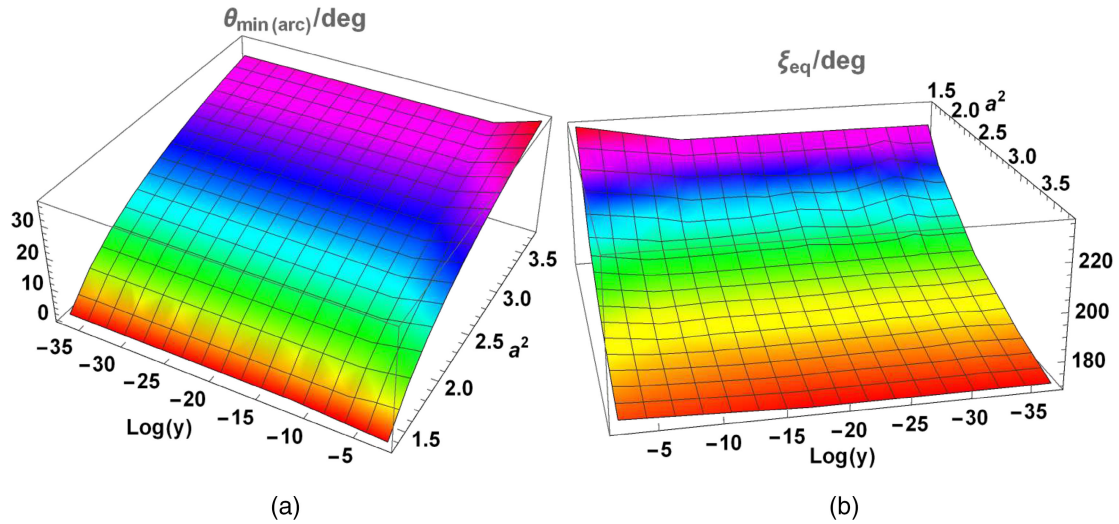


FIG. 31. Dependence of the characteristic angles on the spacetime parameters corresponding to class IVb: (a) the minimum observer's latitude $\theta_{\min(arc)}$ allowing the observer to see the light arc lining the superspinar/NS shadow, as described, e.g., in Fig. 23(b); (b) the maximum angle ξ_{eq} of the arc seen in the equatorial plane [cf. Fig. 29(b)].

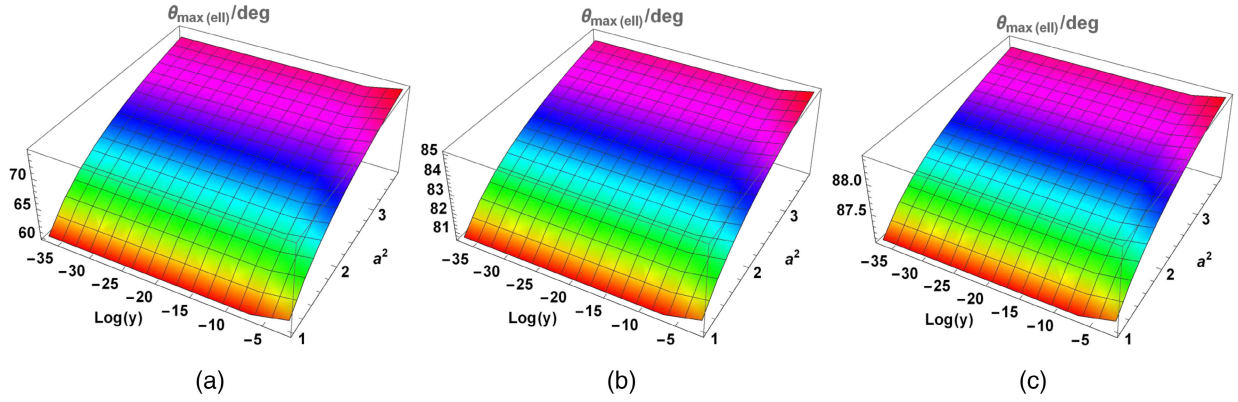


FIG. 32. Dependence of the angle $\theta_{\max(ell)}$, at which the observer sees the elliptical shadow of the superspinar touching the light arc, on the spacetime parameters corresponding to class IV. The superspinar boundary surface is set to $\mathcal{R} = 0.1$ (a), $\mathcal{R} = 0.01$ (b), and $\mathcal{R} = 0.001$ (c).

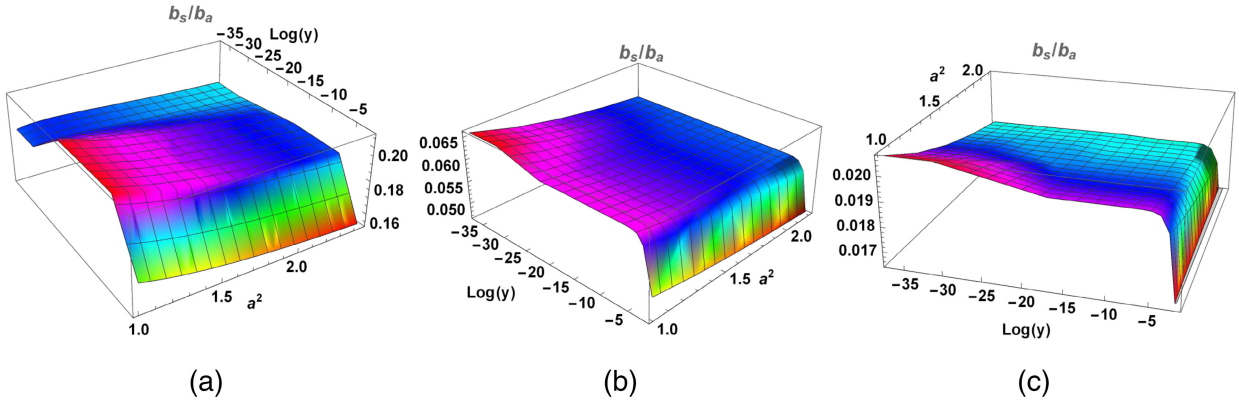


FIG. 33. Dependence of the ratio b_s/b_a as defined in Fig. 25(b). The superspinar boundary surface is set to $\mathcal{R} = 0.1$ (a), $\mathcal{R} = 0.01$ (b), and $\mathcal{R} = 0.001$ (c). It can be seen that the smaller the value of the radius $\mathcal{R}$ of the superspinar surface, the narrower its elliptical shadow relative to the appropriate light arc.

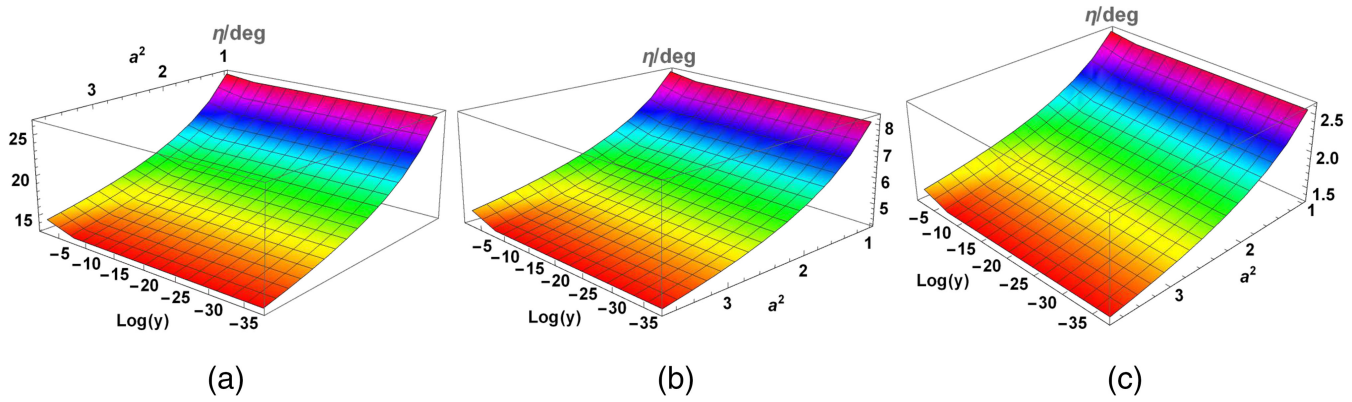
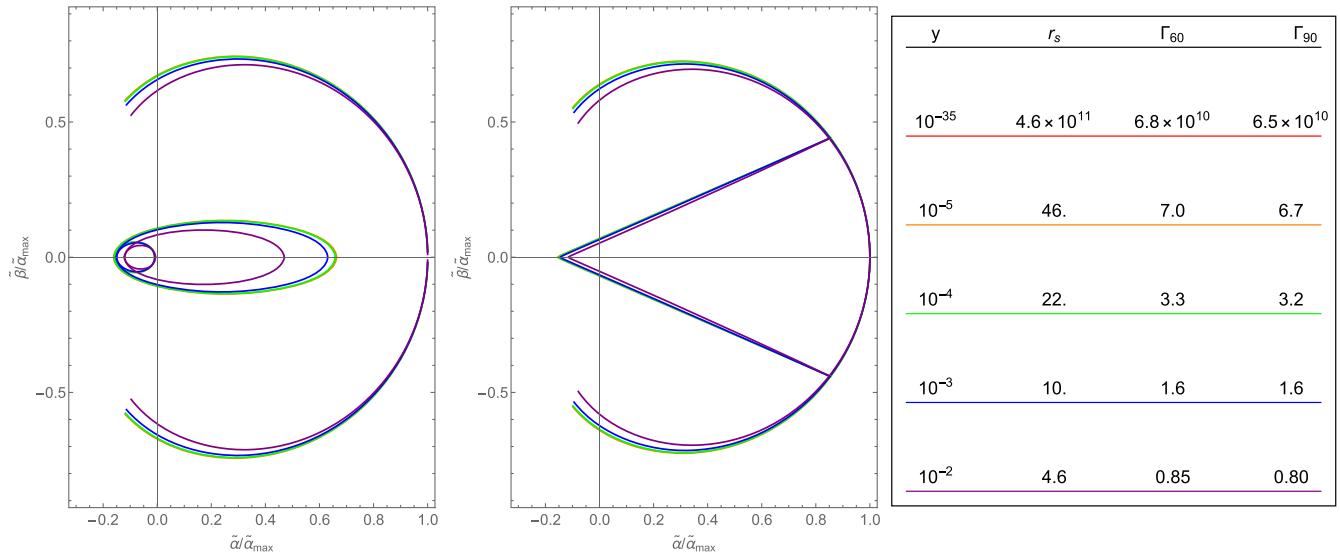


FIG. 34. Dependence of the angle η, which defines the maximum shadow of the superspinar observed in the equatorial plane, on the spacetime parameters corresponding to class IV. The superspinar boundary surface is set to $\mathcal{R} = 0.1$ (a), $\mathcal{R} = 0.01$ (b), and $\mathcal{R} = 0.001$ (c).



y	r_s	Γ_{60}	Γ_{90}
10^{-35}	4.6×10^{11}	6.8×10^{10}	6.5×10^{10}
10^{-5}	46.	7.0	6.7
10^{-4}	22.	3.3	3.2
10^{-3}	10.	1.6	1.6
10^{-2}	4.6	0.85	0.80

FIG. 35. Superspinar vs NS shadows for fixed spin parameter $a^2 = 1.2$ and different values of the cosmological parameter y as seen by an observer located at the static radius $r_s = y^{-1/3}$ with latitude $\theta_o = 60°$ (left) and at the equatorial plane $\theta_o = 90°$ (middle). To clearly show the influence of the cosmological parameter on the shape of the shadow, the images are resized by the multiplicative factor $\Gamma = 1/\tilde{\alpha}_{\max}$, so that the modified dimensionless coordinate $\tilde{\alpha}/\tilde{\alpha}_{\max}$ of their left end point is normalized to 1. The shadows contain the arcs determined by photons with motion constants $X_{\rm SPO}(r_{\rm SPO} > 0), q_{\rm SPO}(r_{\rm SPO} > 0)$, which enclose the elliptical (left figure) or wedge-shaped (middle figure) shadows of the superspinar with surface at radius $\mathcal{R} = 0.1$. In the left figure, for comparison with the naked singularity case, we show the images of the dark spots, represented by smaller inner ellipses, through which the light can enter the region of negative radii, enclosed by photons with motion constants $X_{\rm SPO}(r_{\rm SPO} < 0), q_{\rm SPO}(r_{\rm SPO} < 0)$. In the case $\theta_o = 90°$ these spots disappear. Note that the red and orange curves almost overlap, so that for realistic values of the cosmological parameter $y \lesssim 10^{-22}$, where the upper limit corresponds to the most massive objects observed so far, these shadows are indistinguishable from those in the case of pure Kerr spacetime. The right panel gives the corresponding values of y, r_s and the factor Γ by which each corresponding image should be resized to obtain the real angular coordinate $\tilde{\alpha}, \tilde{\beta}$ in radians.

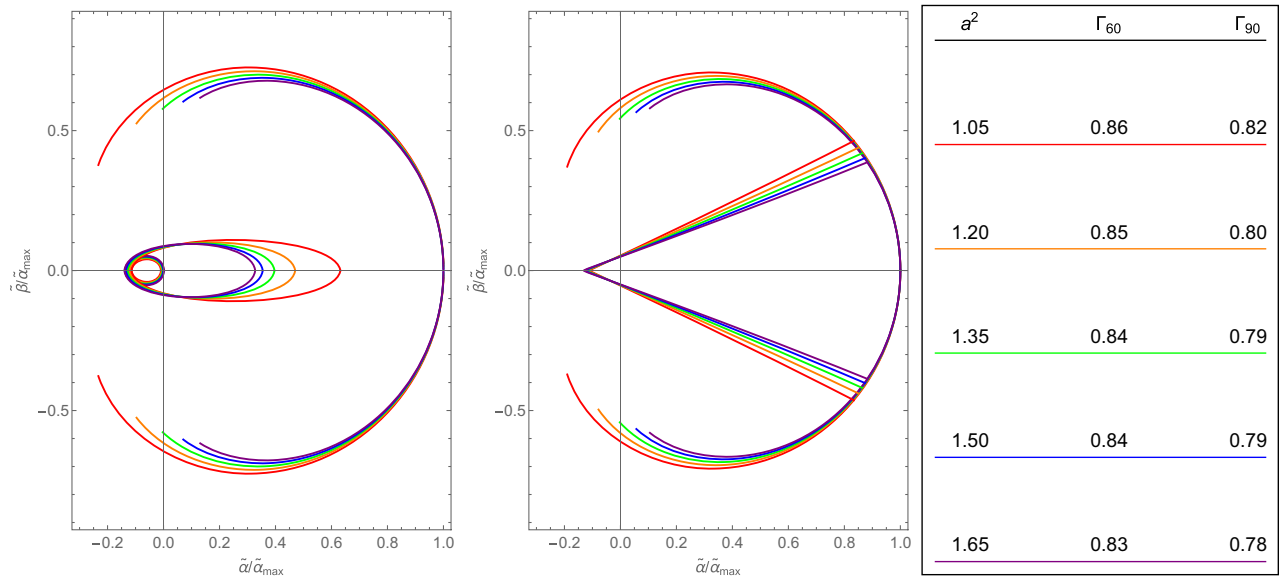


FIG. 36. Superspinar vs NS shadows for fixed cosmological parameter $y = 0.01$ and different values of the spin parameter a^2 as seen by an observer located at the static radius $r_s = y^{-1/3} = 4.6$ with latitude $\theta_o = 60°$ (left) and in the equatorial plane $\theta_o = 90°$ (middle). The description of the shadows is the same as in the previous figure. In order to clearly display the influence of the spin parameter on the shape of the shadow, the images are resized by multiplicative factor $\Gamma = 1/\tilde{\alpha}_{\max}$ so that the modified dimensionless coordinate $\tilde{\alpha}/\tilde{\alpha}_{\max}$ of their left end point is normalized to 1. The right panel gives the corresponding values of a^2 and the factor Γ by which each image should be resized to obtain the real angular coordinate $\tilde{\alpha}, \tilde{\beta}$ in radians. The first three values of the parameter a^2 correspond to spacetimes with polar SPOs, while the remaining two values correspond to spacetimes without polar SPOs. The middle value is close to the marginal value $a^2_{\max(pol)}(y = 0.01) \approx 1.355$, where $a^2_{\max(pol)}(y)$ denotes the inverse of the function $y_{\max(pol)}(a^2)$ defined in (65).

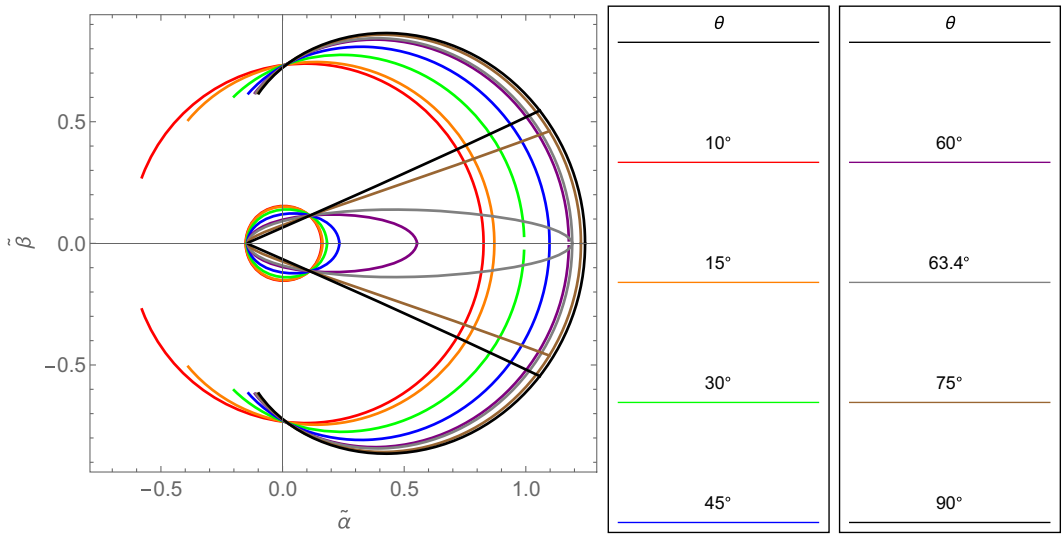


FIG. 37. Superspinar shadows for fixed cosmological parameters $y = 0.01$, $a^2 = 1.2$, corresponding to spacetime IVa with polar SPOs and for different values of the observer's latitude θ, as seen by an observer located at the static radius $r_s = 0.01^{-1/3} = 4.6$. The description of the shadows is the same as in the previous figure. The first value of the observer latitude used is close to the critical value $\theta_{\max(circ)} = 9.04°$.

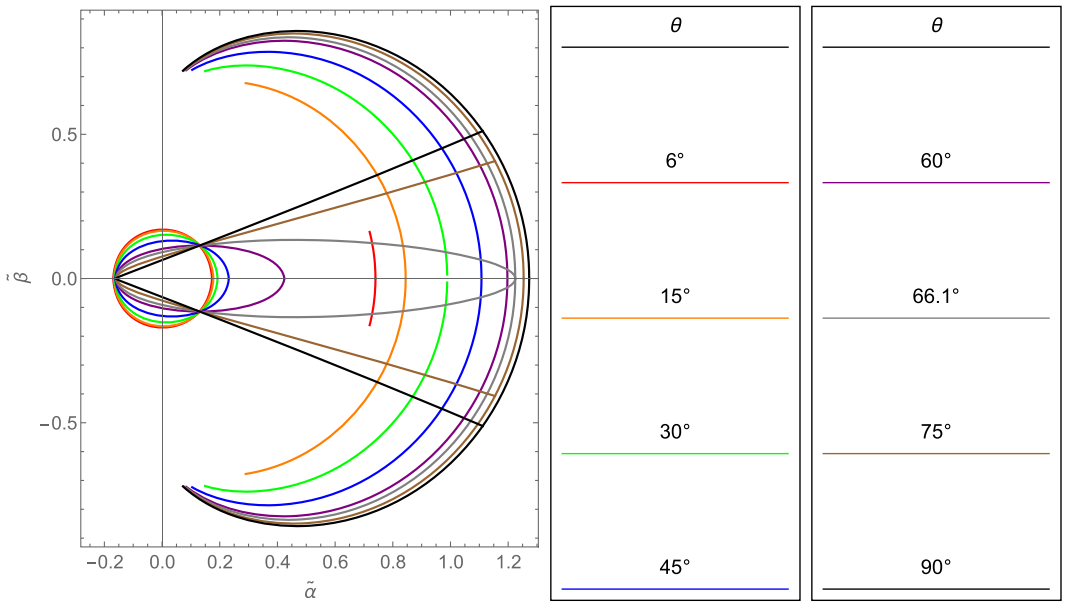


FIG. 38. Superspinar shadows for fixed cosmological parameters $y = 0.01$, $a^2 = 1.5$, corresponding to spacetime IVb without polar SPOs, and for different values of the observer latitude θ as seen by an observer located at the static radius $r_s = 0.01^{-1/3} = 4.6$. The first value of the observer latitude used is close to the critical value $\theta_{\min(arc)} = 5.85°$.

the static radius. The spacetime parameters in Fig. 37 correspond to class IVa, while in Fig. 38 they correspond to class IVb. It can be seen that while the dependence on the spin parameter is significant, the influence of the astrophysically acceptable values of the cosmological parameter is quite negligible.

VI. CONCLUDING REMARKS

In the present paper we have performed an analysis allowing us to construct the light escape cones and related shadows in the Kerr–de Sitter superspinar (NS) spacetimes. The main results obtained can be summarized as follows.

(1) For emitters located under the radii of the equatorial circular photon orbit at $r = r^-_{ph}$, there exist a wide variety of shapes of light cones containing the photons emitted to bound orbits, which we call trapped cones. In the 3D depiction they can take the shapes of spherical belts [see Figs. 13(a) and 16(a)], spherical biangles [see Figs. 14(a), 16(b), 16(c), 17(c), 18, and 20], or spherical canopy with an elliptical base (Fig. 24), in dependence on both the radial and latitudinal position coordinates. The crucial role plays the relative radial position with respect to the outer unstable polar SPO at radii $r^-_{\rm pol}$, if it exists, and the radius of the equatorial circular photon orbit r^-_{ph}. The trapped cones are relevant for the study of self-illumination and self-eclipse effects in the presence of radiating matter, e.g., in an accretion disk.

(2) In Sec. IV C 1 we introduced a subdivision of the complementary cones to the LECs, i.e., the trapped cones, the cones of light escaping to the other infinity, the repelled cones, and the engrossed cones, which for $r_e < r^-_{ph}$ are separated from the LECs by photons emitted toward the unstable SPOs. For emitters at $r_e > r^-_{ph}$ there are no trapped cones.

(3) In the ergoregion there are photon orbits with negative energy. In the case of spacetimes with DRB of the radial photon motion, there is an inner ergoregion, where the corresponding light cones are depicted in Figs. 12–15 in blue, and the outer ergoregion, where the corresponding light cones are shown in Fig. 22 in green.

(4) The 3D representations of the LECs can also be regarded as the star globes showing the celestial sky and the superspinar (NS) shadows viewed from the point where the emitter (observer) resides. The light gray areas, originally representing the trapped cones, should then be understood as the dark sky on the observer's celestial sphere, since no photons from distant radii can come from these directions. Similarly, the red areas on the globe, originally representing the engrossed cones, have the meaning of the superspinar shadow.

(5) For the astrophysically relevant case of a distant observer at $r_o > r^-_{ph}$, the superspinar (NS) shadow consists of the dark spot as described above, the silhouette of the ring singularity itself, which is

one-dimensional and hence not observable, and an arc or circle on the observer's celestial sky, along which the images of distant stars are distorted, as the photons coming from such directions had to circle many times around the appropriate spherical orbits.

The appearance of the arc is strongly affected by the spacetime parameters and the observer's latitude θ_o. In the spacetimes with the polar SPOs and for observers on and near the spin axis, the shadow constitutes from the spot and a circle, which separates for some critical angle $\theta_{\max(circ)}$ into an arc [cf. Figs. 21(a) and 21(b)]. Such separating angle exists even for trapped cones [cf., e.g., Figs. 14(a), 14(b), and 15(a)].

In the spacetimes without the polar SPOs there is neither a circle nor an arc seen from the spin axis or its vicinity, until a minimum angle $\theta_{\min(arc)}$ is attained [cf. Figs. 23(b) and 23(c)]. For larger latitude $\theta_o > \theta_{\min(arc)}$ an arc appears.

(6) We have introduced observable parameters of the shadows (see Fig. 25), the measurement of which can in principle be used to identify the observed object and to determine its real parameters. We have studied the influence of the spacetime parameters on these observables. We found some influence of the cosmological constant on the shape of the shadows, but the influence of the spin is clearly dominant.

For instance, an arc indicates the presence of a naked singularity whose central angle ξ, defined in Fig. 25(a), is highly sensitive on the spin. From Fig. 29 it follows that for $y < y_{\max(pol)}(a^2)$, i.e., in the spacetimes with the polar SPOs, there is some minimum value of the central angle for any latitude of the observation, under which the central angle cannot decrease. At least, the detection of an arc with $\xi \to 360°$ points to a small latitude (high inclination) of the observation and gives a strong upper limit on the spin. On the other hand, under the same conditions, the observation of an arc with $\xi \to 0°$ indicates the spin $y > y_{\max(pol)}(a^2)$. Moreover, the latitude of the observation can be inferred from the shape of the superspinar shadow (elliptical or wedge shaped). Of course, a practical measurement of such an arc from the distortion of the celestial background is rather a task for advanced devices in the future.

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